

→ Glucose meter

→ Wireless Charger

→ Transformer

→ Satellites

→ Motor

→ Laptop / Smart Watch

→ IC's

→ Bullet train

1) Transmission line

2) Vector Calculus

3) Static Electric Field / Static Magnetic Field

4) Maxwell's equation

Gauss's law:-

$$\phi = \frac{Q}{\epsilon_0}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{D.F})$$

$$\oint \vec{E} \cdot \hat{n} da = \frac{Q_{enc}}{\epsilon_0} \quad (\text{I.F})$$

Gauss's law for magnetism:-

$$\oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{I.F})$$

$$\nabla \cdot \vec{B} = 0 \quad (\text{D.F})$$

Ampere's law:-

$$\nabla \times \vec{B} = \mu_0 \vec{J} \quad (\text{D.F})$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I + \epsilon_0 \mu_0 \frac{d\phi}{dt} \quad (\text{I.F})$$

Faraday's law:-

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\phi_B}{dt} \quad (\text{I.F})$$

$$\nabla \times \vec{E} = - \frac{d\vec{B}}{dt} \quad (\text{D.F})$$

Ch#1

Scalar

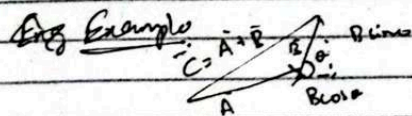
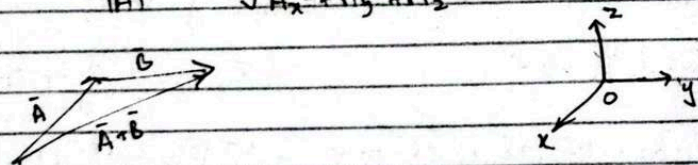
- Magnitude
- One dimensional
- Scalar ^{can be} divided by scalar ~~into~~
- Time, mass, length

Vector

- Magnitude + Direction
- Multi dimensional
- Vector cannot be divided by another vector
- Velocity, force, weight

Basics of Vector:-

- $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
- $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$ (mag)
- $\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{A_x \hat{i} + A_y \hat{j} + A_z \hat{k}}{\sqrt{A_x^2 + A_y^2 + A_z^2}} = \hat{i}_A$ (unit vector)



Using pythagoras theorem,

$$C^2 = (A + B \cos \theta)^2 + (B \sin \theta)^2$$

$$C^2 = A^2 + B^2 \cos^2 \theta + 2AB \cos \theta + B^2 \sin^2 \theta$$

$$C^2 = A^2 + B^2 + 2AB \cos \theta$$

Dot Product

$$C = \vec{A} \cdot \vec{B}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$$

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta \quad (\text{scalar resultant})$$

$$= A_x B_x + A_y B_y + A_z B_z$$

Cross Product:-

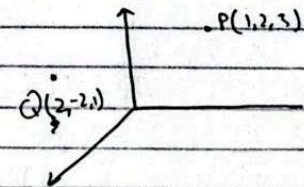
$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad (\text{vector resultant})$$

Example:-

$$\vec{P} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{Q} = 2\hat{i} - 2\hat{j} + \hat{k}$$



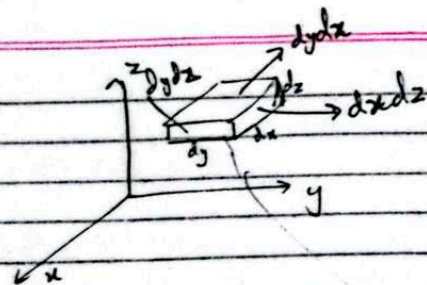
$$\vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= 2\hat{i} - 2\hat{j} + \hat{k} - \hat{i} - 2\hat{j} - 3\hat{k}$$

$$\vec{PQ} = \hat{i} - 4\hat{j} - 2\hat{k}$$

$$|\vec{PQ}| = \sqrt{1^2 + (-4)^2 + (-2)^2}$$

$$= \sqrt{21}$$



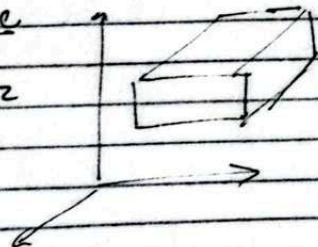
$$dV = dx dy dz$$

$$|dL| = \sqrt{dx^2 + dy^2 + dz^2}$$

Example

$$dV = dx dy dz$$

$$\approx 0.06$$



$P(1,1,3)$

$$dx = 0.1$$

$$dy = 0.2$$

$$dz = 0.3$$

Eg

Specify the unit vector from the origin towards the point $F(2,-2,1)$

$$\vec{F} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$|\vec{F}| = \sqrt{2^2 + (-2)^2 + 1^2} = \sqrt{9} = 3$$

$$\hat{F} = \frac{\vec{F}}{|\vec{F}|} = \frac{2\hat{i} - 2\hat{j} + \hat{k}}{3}$$

$$\hat{F} = \frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k}$$

Eg

$U(-1,2,1)$, $L(3,-3,0)$, $S(-2,-3,-4)$

$\vec{R}_{UL} = ?$, $\vec{R}_{UL} + \vec{R}_{US} = ?$, $|\vec{R}_{UL}| = ?$, $|\vec{R}_{US}| = ?$

unit vector of $\vec{R}_{UL} = ?$

$$\vec{R}_{UL} = -\hat{i} + 3\hat{j} + \hat{k}$$

$$|\vec{R}_{UL}| = \sqrt{(-1)^2 + (3)^2 + (1)^2} = \sqrt{11}$$

$$\vec{R}_{US} = 3\hat{i} - 3\hat{j}$$

$$\vec{R}_{UL} = \vec{R}_L - \vec{R}_U$$

$$= 3\hat{i} - 3\hat{j} + \hat{i} - 2\hat{j} - \hat{k}$$

$$= 4\hat{i} - 5\hat{j} - \hat{k}$$

$$\vec{R}_S = -2\hat{i} - 3\hat{j} - 4\hat{k}$$

$$|\vec{R}_S| = \sqrt{(-2)^2 + (-3)^2 + (-4)^2}$$

$$= \sqrt{29}$$

$$\vec{R}_{US} = \vec{R}_S - \vec{R}_U$$

$$= -2\hat{i} - 3\hat{j} - 4\hat{k} + \hat{i} - 2\hat{j} - \hat{k}$$

$$= -\hat{i} - 5\hat{j} - 5\hat{k}$$

$$\vec{R}_{UL} + \vec{R}_{US} = 4\hat{i} - 5\hat{j} - \hat{k} - \hat{i} - 5\hat{j} - 5\hat{k}$$

$$= 3\hat{i} - 10\hat{j} - 6\hat{k}$$

$$|\vec{R}_{UL}| = \sqrt{4^2 + (-5)^2 + (-1)^2}$$

$$= \sqrt{42}$$

$$\hat{R}_{UL} = \frac{\vec{R}_{UL}}{|\vec{R}_{UL}|}$$

$$= \frac{-\hat{i} - 5\hat{j} - 5\hat{k}}{\sqrt{42}}$$

$$= -\frac{1}{\sqrt{42}}\hat{i} - \frac{5}{\sqrt{42}}\hat{j} - \frac{5}{\sqrt{42}}\hat{k}$$

$$\rho(\text{row}) = \text{radius} \quad s = \frac{r}{H} \quad C = \frac{R}{H} \quad T = \frac{P}{B}$$

$$s = \frac{P}{H} \quad C = \frac{R}{H} \quad T = \frac{P}{B}$$

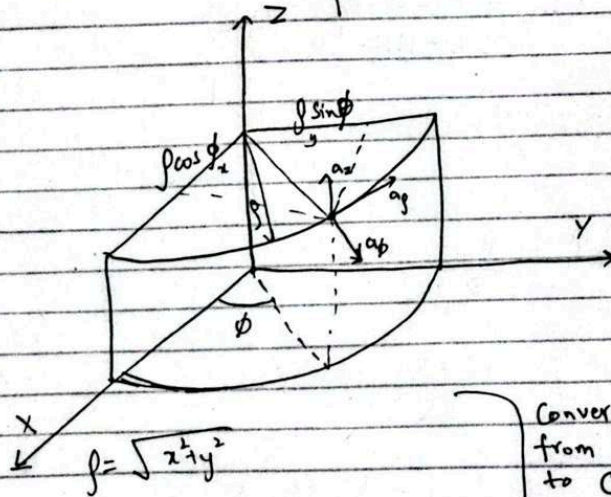
Cylindrical Coordinate System

Real world applications:-

- 1) Electromagnetics
- 1) Fluid dynamics
- 1) Electromagnetic field in wires
- 1) Flow in pipes
- 1) Heat conduction in cylinders

Cylindrical Coordinate System | Rectangular Coordinate System

- | | |
|---|---|
| <ul style="list-style-type: none"> 1) Defined by ρ, ϕ and z 1) ρ (radial distance from z-axis) 1) ϕ (angle it makes in the xy plane) 1) z (distance along z axis) | <ul style="list-style-type: none"> 1) Defined by x, y and z axis 1) Surfaces of constant ρ are centered along z-axis 1) Surfaces of constant ϕ are planes passing through z-axis 1) Surfaces of constant z are horizontal planes |
|---|---|



$$\tan \phi = \frac{y}{x} \Rightarrow \phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$z = z$$

Conversion from RCS to CCS

Conversion from CCS to RCS

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$

Unit Vectors

$$a_\rho, a_\phi, a_z \text{ (CCS)} \quad a_x, a_y, a_z \text{ (RCS)}$$

Transforming unit Vector

	Dot Product		
	a_ρ	a_ϕ	a_z
a_x	$\cos \phi$	$-\sin \phi$	0
a_y	$\sin \phi$	$\cos \phi$	0
a_z	0	0	1

$$A_{\text{rect}} = A_\rho a_\rho + A_\phi a_\phi + A_z a_z$$

$$A_{\text{rect}} = A_x a_x + A_y a_y + A_z a_z$$

⇒ To find any desired component of a vector, we take the dot product of the vector and a unit vector in desired direction.

$$A_\rho = A_{\text{rect}} \cdot a_\rho$$

$$A_\phi = A_{\text{rect}} \cdot a_\phi$$

$$A_z = A_z$$

$$A_p = A_{rect} \cdot a_p = (A_x a_x + A_y a_y + A_z a_z) \cdot a_p$$

$$= A_x (a_x \cdot a_p) + A_y (a_y \cdot a_p) + A_z (a_z \cdot a_p)$$

$$= A_x \cos \phi + A_y \sin \phi + A_z (0)$$

$$A_p = A_x \cos \phi + A_y \sin \phi$$

$$\phi = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

$$A_p = A_{rect} \cdot a_p = (A_x a_x + A_y a_y + A_z a_z) \cdot a_p$$

$$= A_x (a_x \cdot a_p) + A_y (a_y \cdot a_p) + A_z (a_z \cdot a_p)$$

$$= A_x (-\sin \phi) + A_y \cos \phi + A_z (0)$$

$$A_p = -A_x \sin \phi + A_y \cos \phi$$

$$A_z = A_z$$

$$Q) B = y a_x - x a_y + z a_z$$

$$B_{cy} = B_p a_p + B_\phi a_\phi + B_z a_z$$

$$B_p = B \cdot a_p$$

$$B_p = y \cos \phi - x \sin \phi$$

$$B_\phi = B \cdot a_\phi$$

$$B_\phi = -y \sin \phi - x \cos \phi$$

$$B_z = B_z$$

$$D(1.5) a) C_{cy} = C_p a_p + C_\phi a_\phi + C_z a_z = 4.4 a_p + 11.5 a_\phi + 2 a_z$$

$$C_x = C_p \cos \phi = 4.4 \cos(-115) = -1.86$$

$$C_y = C_p \sin \phi = 4.4 \sin(-115) = -3.99$$

$$C_z = C_z = 2$$

$$C_{rect} = C_x a_x + C_y a_y + C_z a_z = -1.86 a_x + 3.99 a_y + 2 a_z$$

$$b) D_{rect} = D_x a_x + D_y a_y + D_z a_z = -3.1 a_x + 2.6 a_y + 3 a_z$$

$$D_p = \sqrt{D_x^2 + D_y^2} = \sqrt{(-3.1)^2 + (2.6)^2} = 4.05$$

$$\theta_p = \tan^{-1} \left(\frac{D_y}{D_x} \right) = \tan^{-1} \left(\frac{2.6}{-3.1} \right) = 39.99^\circ$$

$$D_\phi = 180^\circ - \theta = 180^\circ - 39.99^\circ = 140.01^\circ$$

$$D_z = D_z = -3$$

$$D_{cy} = D_p a_p + D_\phi a_\phi + D_z a_z = 4.05 a_p + 140.01 a_\phi + 3 a_z$$

$$c) \bar{r}_{co} = \overline{OD}_{rect} - \overline{OC}_{rect}$$

$$= -3.1 a_x + 2.6 a_y + 3 a_z + 1.86 a_x + 3.99 a_y - 2 a_z$$

$$= -1.24 a_x + 6.59 a_y - 5 a_z$$

$$|\bar{r}_{co}| = \sqrt{(-1.24)^2 + (6.59)^2 + (-5)^2}$$

$$= 8.36$$

D1.6) a) $F_{\text{rect}} = 10a_x - 8a_y + 6a_z$ at $(10, -8, 6)$

$$F_g = \sqrt{10^2 + (-8)^2} = 12.81$$

$$F_\phi = \tan^{-1}\left(\frac{-8}{10}\right) = -38.66^\circ$$

$$F_z = F_z = 6$$

$$F_{\text{cyl}} = 12.81a_\rho + 38.66a_\phi + 6a_z$$

b) $G_{\text{rect}} = (2x+y)a_x + (y-4x)a_y$ at (ρ, ϕ, z)

$$G_\rho = G_{\text{rect}} \cdot a_\rho = (2x+y)a_x + (y-4x)a_y \cdot a_\rho$$

$$\begin{aligned} &= (2x+y)\cos\phi - (y-4x)\sin\phi \\ &= 2x\cos\phi + y\cos\phi - y\sin\phi + 4x\sin\phi \\ &= 2(\rho\cos\phi)\cos\phi + (\rho\sin\phi)\cos\phi - (\rho\sin\phi)\sin\phi \\ &\quad + 4(\rho\cos\phi)\sin\phi \\ &= 2\rho\cos^2\phi + \rho\sin\phi\cos\phi - \rho\sin^2\phi \\ &\quad + 4\rho\cos\phi\sin\phi \\ &= 2\rho\cos^2\phi - \rho\sin^2\phi + 5\rho\sin\phi\cos\phi \end{aligned}$$

$$\begin{aligned} G_\phi &= -(2x+y)\sin\phi + (y-4x)\cos\phi \\ &= -2x\sin\phi - y\sin\phi - y\cos\phi + 4x\cos\phi \\ &= -2\rho\cos\phi\sin\phi - \rho\sin^2\phi - \rho\cos\phi\sin\phi + 4\rho\cos^2\phi \\ &= 4\rho\cos^2\phi - \rho\sin^2\phi - 3\rho\sin\phi\cos\phi \end{aligned}$$

$$G_z = 0$$

$$G_{\text{cyl}} = (2\rho\cos^2\phi - \rho\sin^2\phi + 5\rho\sin\phi\cos\phi)a_\rho + (4\rho\cos^2\phi - \rho\sin^2\phi - 3\rho\sin\phi\cos\phi)a_\phi$$

c) $H = 20a_\rho - 10a_\phi + 3a_z$ at $P(5, 2, -1)$

$$\phi = \tan^{-1}\left(\frac{2}{5}\right) = 21.80^\circ$$

$$\begin{aligned} H_x &= H \cdot a_x = 20\cos\phi - 10\sin\phi \\ &= 20\cos(21.80^\circ) - 10\sin(21.80^\circ) \\ &= 27.28 \end{aligned}$$

$$\begin{aligned} H_y &= H \cdot a_y = 20\sin\phi - 10\cos\phi \\ &= 20\sin(21.80^\circ) - 10\cos(21.80^\circ) \\ &= -1.858 \end{aligned}$$

$$H_z = 3$$

(a) $F = 10a_x - 8a_y + 6a_z$ at $P(10, -8, 6)$

$$\begin{aligned} F_\rho &= F \cdot a_\rho = 10\cos\phi - 8\sin\phi \\ &= 10\cos(-38.66^\circ) - 8\sin(-38.66^\circ) \\ &= 12.81 \end{aligned}$$

$$\begin{aligned} F_\phi &= F \cdot a_\phi = -10\sin\phi - 8\cos\phi \\ &= -10\sin(-38.66^\circ) - 8\cos(-38.66^\circ) \\ &= 0 \end{aligned}$$

$$F_z = 6$$

$$F_{\text{cyl}} = 12.81a_\rho + 6a_z$$



Spherical Coordinate System

$\Rightarrow r, \phi, \theta \Rightarrow$

$r \Rightarrow$ distance from origin to point

θ (polar angle) $\Rightarrow$ angle b/w z-axis and line connecting to origin

ϕ (Azimuthal angle) $\Rightarrow$ angle b/w x-axis and projection of line connecting the origin to point onto the xy-plane.

RCS - 3 planes

CCS - 2 plane, 1 curve

SCS - A sphere, cone, plane

$\hookrightarrow$ Unit vectors: a_r, a_θ, a_ϕ

$a_r \rightarrow$ radial unit vector, directed outward from origin, normal to the sphere

$a_\theta \rightarrow$ unit vector, tangent to the sphere and pointing in the direction of increasing θ , normal to the conical sphere

$a_\phi \rightarrow$ unit vector, tangent to the sphere and cone, pointing in direction of increasing ϕ

$$x = r \sin \theta \cos \phi$$

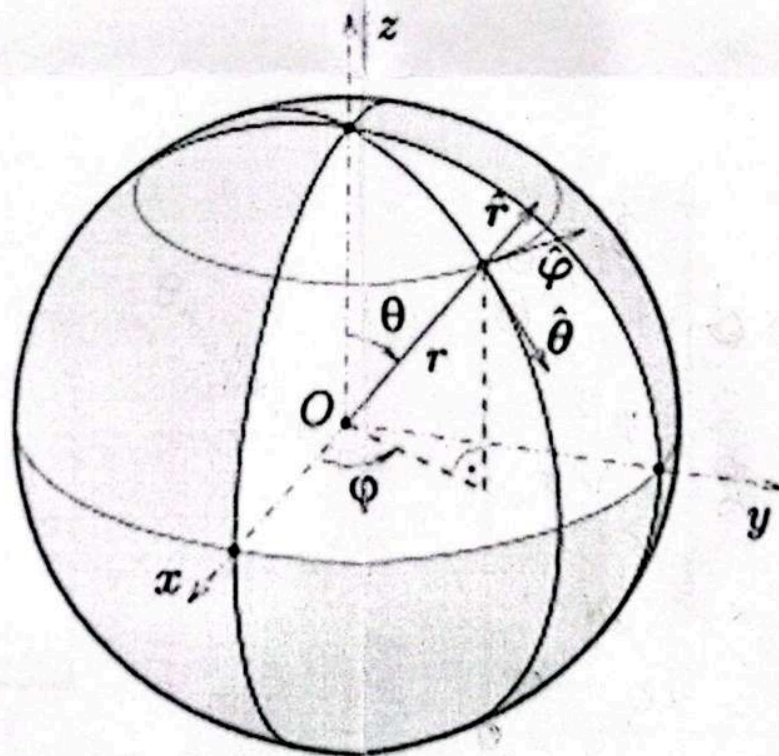
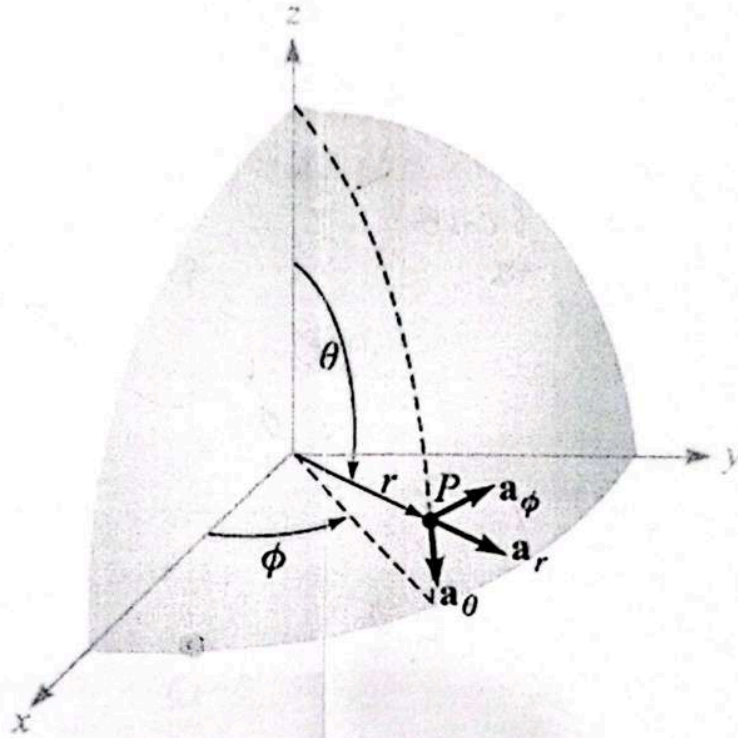
$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

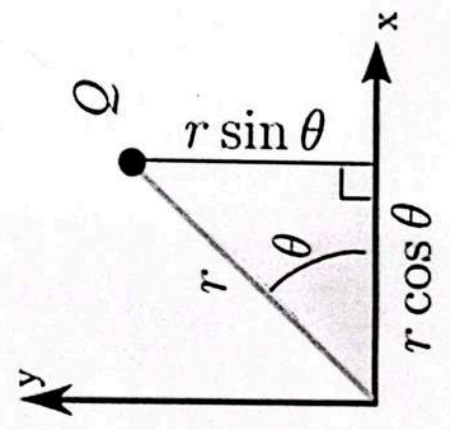
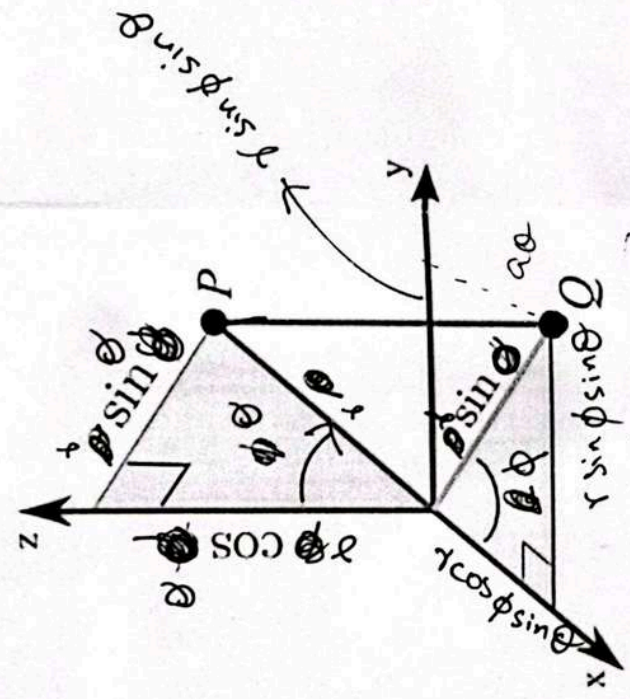
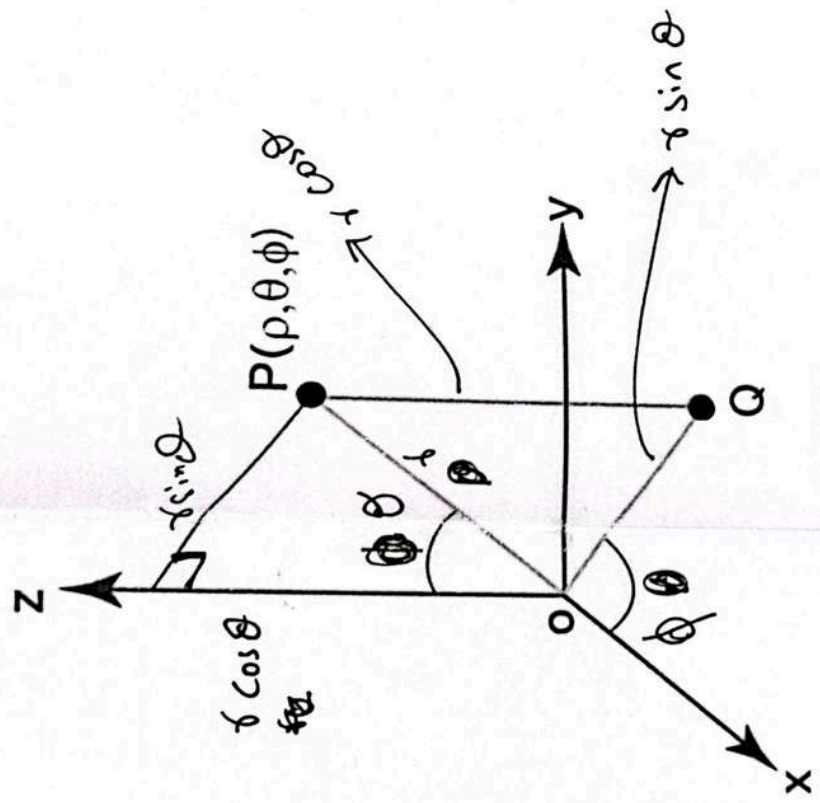
Spherical Coordinate System

Pg 1

SCS - r, ϕ, θ



Pg 2



Coordinate System

in origin to point
angle b/w z-axis and
x-axis and

Transformation from RCS to SCS

$$(x, y, z) \longrightarrow (r, \theta, \phi)$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$z = r \cos \theta$$

$$\cos \theta = \frac{z}{r}$$

$$\theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$r = \sqrt{x^2 + y^2 + z^2}; (r \geq 0)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$; 0 \leq \phi \leq 360^\circ$$

$$; (0 \leq \theta \leq 180^\circ)$$

$r \rightarrow$ radius

$\theta \rightarrow$ w.r.t z-axis

$\phi \rightarrow$ projection into xy plane

Transformation of Vectors:-

The transformation of vectors requires determining the dot products of spherical unit vectors with rectangular unit vectors.

Purpose of dot Product:-

To find the component of a vector in the direction of a new unit vector (from SCS), we take the dot product of original vectors (from RCS) with new vector.

Mathematical Representation:-

$$\vec{A}_{res} = A_{res} = A_x \vec{a}_x + A_y \vec{a}_y + A_z \vec{a}_z$$

In spherical coordinate system,

$$\vec{A}_{scs} = A_r \vec{a}_r + A_\theta \vec{a}_\theta + A_\phi \vec{a}_\phi$$

The components of the vector $\vec{A}_{res}$ along spherical coordinate axis

$$A_x = \vec{A}_{res} \cdot \vec{a}_x = A_x (\vec{a}_x \cdot \vec{a}_x) + A_y (\vec{a}_y \cdot \vec{a}_x) + A_z (\vec{a}_z \cdot \vec{a}_x)$$

$$\vec{a}_x \cdot \vec{a}_x = \vec{a}_x \cdot \vec{a}_x = \cos \phi \sin \theta$$

$$\vec{a}_y \cdot \vec{a}_x = \vec{a}_y \cdot \vec{a}_y = \sin \phi \sin \theta$$

$$\vec{a}_z \cdot \vec{a}_x = \vec{a}_x \cdot \vec{a}_z = \cos \theta$$

$$A_\phi = A_{res} \cdot \vec{a}_\phi$$

$$A_\theta = A_{res} \cdot \vec{a}_\theta$$

$$A_\theta = \vec{A}_{res} \cdot \vec{a}_\theta = A_x (\vec{a}_x \cdot \vec{a}_\theta) + A_y (\vec{a}_y \cdot \vec{a}_\theta) + A_z (\vec{a}_z \cdot \vec{a}_\theta)$$

$$\vec{a}_x \cdot \vec{a}_\theta = \vec{a}_\theta \cdot \vec{a}_x = \cos \theta \cos \phi$$

$$\vec{a}_y \cdot \vec{a}_\theta = \vec{a}_\theta \cdot \vec{a}_y = \cos \theta \sin \phi$$

$$\vec{a}_z \cdot \vec{a}_\theta = \vec{a}_\theta \cdot \vec{a}_z = -\sin \theta$$

$$A_\phi = \vec{A}_{res} \cdot \vec{a}_\phi = A_x (\vec{a}_x \cdot \vec{a}_\phi) + A_y (\vec{a}_y \cdot \vec{a}_\phi) + A_z (\vec{a}_z \cdot \vec{a}_\phi)$$

$$\vec{a}_x \cdot \vec{a}_\phi = \vec{a}_\phi \cdot \vec{a}_x = -\sin \phi$$

$$\vec{a}_y \cdot \vec{a}_\phi = \vec{a}_\phi \cdot \vec{a}_y = +\cos \phi$$

$$\vec{a}_z \cdot \vec{a}_\phi = \vec{a}_\phi \cdot \vec{a}_z = 0$$

	$\vec{a}_r$	$\vec{a}_\theta$	$\vec{a}_\phi$
$\vec{a}_x$	$\sin \theta \cos \phi$	$\cos \theta \cos \phi$	$-\sin \phi$
$\vec{a}_y$	$\sin \theta \sin \phi$	$\cos \theta \sin \phi$	$+\cos \phi$
$\vec{a}_z$	$\cos \theta$	$-\sin \theta$	0

$$A_r = A_x \cos \phi \sin \theta + A_y \sin \phi \sin \theta + A_z \cos \theta$$

$$A_\theta = A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi + A_z \sin \theta$$

$$A_\phi = -A_x \sin \phi + A_y \cos \phi$$

$$\vec{A}_{scs} = (A_x \cos \phi \sin \theta + A_y \sin \phi \sin \theta + A_z \cos \theta) \vec{a}_r + (A_x \cos \theta \cos \phi + A_y \cos \theta \sin \phi + A_z \sin \theta) \vec{a}_\theta + (-A_x \sin \phi + A_y \cos \phi) \vec{a}_\phi$$

Q) $G_{res} = \frac{xz}{y} \vec{a}_x \rightarrow$ in S_1 SCS

$$G_{scs} = G_r \vec{a}_r + G_\theta \vec{a}_\theta + G_\phi \vec{a}_\phi$$

$$G_r = G \cdot \vec{a}_r = \frac{xz}{y} \vec{a}_x \cdot \vec{a}_r$$

$$= \frac{(r \sin \theta \cos \phi)(r \cos \theta)(\vec{a}_x \cdot \vec{a}_r)}{(r \sin \theta \sin \phi)}$$

$$= \frac{r \cos \theta \cos \phi}{\sin \phi} (\sin \theta \cos \phi)$$

$$= \frac{r \cos \theta \cos^2 \phi \sin \theta}{\sin \phi}$$

$$C_{r\theta} = C_r \cdot a_\theta = \frac{a_\theta}{\sin\phi} r \cos\theta \cos\phi (a_r \cdot a_\theta)$$

$$= \frac{r \cos\theta \cos\phi}{\sin\phi} (\cos\theta \cos\phi)$$

$$= \frac{r \cos^2\theta \cos^2\phi}{\sin\phi}$$

$$C_{r\phi} = C_r \cdot a_\phi = \frac{r \cos\theta \cos\phi}{\sin\phi} (a_r \cdot a_\phi)$$

$$= \frac{r \cos\theta \cos\phi}{\sin\phi} (-\sin\phi)$$

$$= -r \cos\theta \cos\phi$$

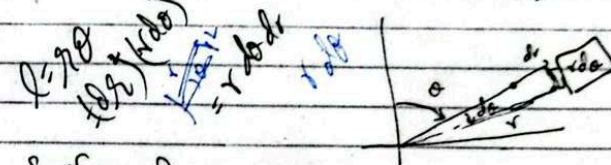
$$\Rightarrow C_{scr} = \left(\frac{r \cos\theta \cos^2\phi \sin\theta}{\sin\phi} \right) a_r + \left(\frac{r \cos^2\theta \cos^2\phi}{\sin\phi} \right) a_\theta$$

$$- (r \cos\theta \cos\phi) a_\phi$$

Differential Elements in SCS

Differential lengths

- Consider increasing r , θ and ϕ by small amounts $\rightarrow dr$, $d\theta$ and $d\phi$ respectively
- $\Rightarrow dr \rightarrow$ along the radial direction (r)
- $\Rightarrow r d\theta \rightarrow$ along the polar direction (θ)
- $\Rightarrow r \sin\theta d\phi \rightarrow$ along the azimuthal direction (ϕ)



Differential Surface Area

- Diff. surface area spanned by r & $\theta = r dr d\theta$
- Diff. surface area spanned by r & $\phi = r \sin\theta dr d\phi$
- Diff. surface area spanned by θ & $\phi = r^2 \sin\theta d\theta d\phi$

Differential Volume: $dr \times r d\theta \times r \sin\theta d\phi = r^2 \sin\theta dr d\theta d\phi$

$$dV = r^2 \sin\theta dr d\theta d\phi$$

$$\int_V dV = \int_V r^2 \sin\theta dr d\theta d\phi = \frac{r^3}{3} \sin\theta d\theta d\phi$$

$$\int_\theta dV = \int_\theta r^2 \sin\theta dr d\theta d\phi = -\frac{r^2}{2} \cos\theta dr d\phi$$

$$\int_\phi dV = \int_\phi r^2 \sin\theta dr d\theta d\phi = \phi r^2 \sin\theta dr d\theta$$

$$\iiint_V dV = \int_0^\pi \int_0^{2\pi} \int_0^1 r^2 \sin\theta dr d\theta d\phi$$

$$= \frac{-r^3}{3} \cos\theta \Big|_0^\pi$$

Q1) C(-3, 2, 1) → SCS

$$C_{res} = -3a_x + 2a_y + a_z$$

$$C_{SCS} = C_r a_r + C_\theta a_\theta + C_\phi a_\phi$$

$$C_r = C_{res} \cdot a_r$$

$$= -3(a_x \cdot a_r) + 2(a_y \cdot a_r) + (a_z \cdot a_r)$$

$$= -3 \sin\theta \cos\phi + 2 \sin\theta \sin\phi + \cos\theta$$

$$C_\theta = C_{res} \cdot a_\theta$$

$$= -3(a_x \cdot a_\theta) + 2(a_y \cdot a_\theta) + (a_z \cdot a_\theta)$$

$$= -3$$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 4 + 1} = \sqrt{14}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{1}{\sqrt{14}}\right) = 74.49^\circ$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2}{-3}\right) = -33^\circ$$

$$= 140.36^\circ$$

Q2) D(r=5, θ=20°, φ=-70°) → RCS

$$x = r \sin\theta \cos\phi$$

$$= 0.5$$

$$y = r \sin\theta \sin\phi$$

$$= -1.66$$

$$z = r \cos\theta$$

$$= 4.89$$

1/2/4/8

Ch#2

$$F = k \frac{q_1 q_2}{r^2}$$

Coulomb's Law

Force between two charged particles is

→ Proportional to the magnitude of each charge (Q_1 and Q_2)

→ Inversely proportional to the square of the distance b/w the two charges (R)

→ Acts along the line joining the two charges

→ Is repulsive if the charges are of same sign and attractive if they are of opposite

→ Q_1 and Q_2 are the magnitudes of the charges.

→ R is the distance b/w the charges

→ K is proportionality constant

$$K = \frac{1}{4\pi\epsilon_0}$$

→ ϵ_0 is permittivity of free space

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Components of Coulomb's Law:-

1) Permittivity of free space (ϵ_0)

Crucial for determining how

electric fields interact in a vacuum.

It is not dimensionless and has units

$$\text{C}^2/\text{Nm}^2 \text{ or } \text{F/m}$$

2) The Coulomb:-

It is the SI unit of electric charge. The charge of a single electron is $-1.602 \times 10^{-19} \text{ C}$.

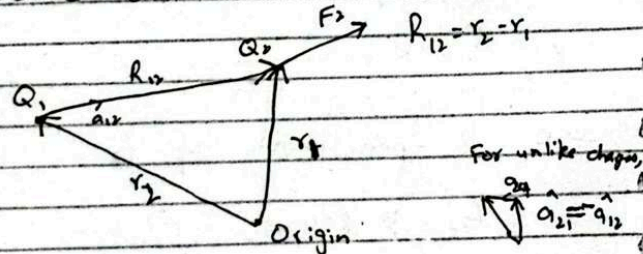
1 Coulomb corresponds to 6×10^{18} electrons

3) Force Magnitude:-

Force b/w two charges of 1C, separated by 1m in a vacuum.
 $9 \times 10^9 \text{ N}$

Vector form of Coulomb's Law

If Q_1 and Q_2 have like signs, the vector force F_{12} on Q_2 is in the same direction as the vector R_{12} .



$$F_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2}$$

$$a_{12} = \frac{R_{12}}{|R_{12}|} = \frac{r_2 - r_1}{|r_2 - r_1|}$$

Electric Field Intensity

Consider a fixed charge Q_1 and test charge Q_2 is moved around Q_1 .

$\rightarrow \hat{a}_{12}$ is the unit vector

pointing from Q_1 to Q_2 . $F_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$

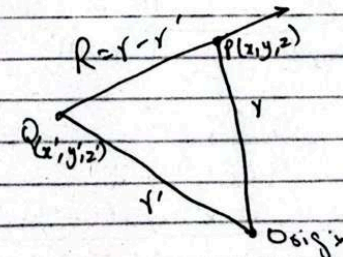
Electric field intensity is the force experienced by a unit positive charge placed at a point in the field due to charge Q_1 .

$$E = \frac{F_{12}}{Q_2} = \frac{Q_1}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

$$F = QE$$

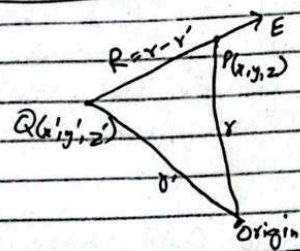
Electric Field of a Point Charge

$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R \rightarrow \hat{a}_R$ is the unit vector pointing from the charge Q to the point where E is measured.



If the charge is placed at the origin of SCS, the unit vector $\hat{a}_R$ becomes the radial unit vector $\hat{a}_R$ and distance R becomes r .

Electric Field in Rectangular Coordinate System



Suppose the charge Q is located at position $\vec{r}' = x'\hat{a}_x + y'\hat{a}_y + z'\hat{a}_z$.

We want to find the electric field intensity at a point $\vec{r} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$.
The distance b/w charge and the point of interest is $\vec{R} = \vec{r} - \vec{r}'$.
Magnitude of $\vec{R}$ is $|\vec{R}| = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$.

$$\begin{aligned} E &= \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_{r-r'} \\ &= \frac{Q}{4\pi\epsilon_0 (|\vec{r}-\vec{r}'|)^2} \frac{(\vec{r}-\vec{r}')}{\sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}} \\ &= \frac{Q}{4\pi\epsilon_0 (|\vec{r}-\vec{r}'|)^3} (\vec{r}-\vec{r}') \\ &= \frac{Q}{4\pi\epsilon_0 (|\vec{r}-\vec{r}'|)^3} ((x-x')\hat{a}_x + (y-y')\hat{a}_y + (z-z')\hat{a}_z) \\ &= \frac{Q}{4\pi\epsilon_0 ((x-x')^2 + (y-y')^2 + (z-z')^2)^{3/2}} ((x-x')\hat{a}_x + (y-y')\hat{a}_y + (z-z')\hat{a}_z) \end{aligned}$$

Eg

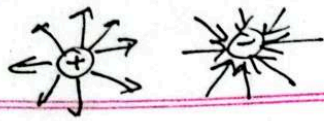
(1) $Q_1 = 4\mu\text{C}$ at $(2, 0, 0)$, $Q_2 = -3\mu\text{C}$ at $(0, 2, 0)$

Calculate total E field at $(1, 1, 0)$

(a) Distance from Q_1 to $(1, 1, 0)$

$$|\vec{r} - \vec{r}_1| = \sqrt{(1-2)^2 + (1-0)^2 + (0-0)^2} = \sqrt{2}$$

$$E_1 = \frac{(4\mu)(-1, 1, 0)}{4\pi\epsilon_0 (2)^{3/2}}$$



Volume Charge Density

$\rho_v \rightarrow$ describes how much charge is distributed in a small volume of space
 C/m^3 (Coulomb per cubic meter)

$$\rho_v = \frac{\text{Total charge}}{\text{Total volume}} = \frac{dQ}{dv}$$

$$dQ = \rho_v dv$$

$$\Rightarrow Q = \int_V \rho_v dv = \iiint_V \rho_v dx dy dz$$

$$\Rightarrow Q = \int_V \rho_v dv = \iiint_V \rho_v r dr d\theta d\phi$$

$$\Rightarrow Q = \int_V \rho_v dv = \iiint_V \rho_v r^2 \sin\theta dr d\theta d\phi$$

$$\Rightarrow dE(r) = \frac{dQ}{4\pi\epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|}$$

$$\Rightarrow E(r) = \int_V \frac{\rho_v(r')}{4\pi\epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|} dv$$

$$E(r) = \int_V \rho_v dv \left(\frac{1}{4\pi\epsilon_0 |r-r'|^2} \right) \left(\frac{r-r'}{|r-r'|} \right)$$

$$= \int_V \frac{\rho_v dv}{4\pi\epsilon_0 r^2} \hat{a}_r$$

$$V = \int_{\text{volume}} \frac{\rho_v dv}{4\pi\epsilon_0 |r-r'|^3}$$

Q) Total charge contained in a 2cm length of an electron beam.

$$\rho_v = -5 \times 10^{-6} \text{ C/m}^3$$

Sol.

$$Q = \int_V \rho_v dv$$

$$= \int_{-0.02}^{0.02} \int_0^{2\pi} \int_0^{0.01} (-5 \times 10^{-6}) r dr d\theta dz$$

$$= \int_{-0.02}^{0.02} \int_0^{2\pi} \left(\frac{2\pi^2 - \theta}{2} \right) d\theta dz (-5 \times 10^{-6})$$

$$= 2\pi^2 (-5 \times 10^{-6}) \int_{-0.02}^{0.02} (1) dz$$

$$= -10\pi^2 \times 10^{-6} (2)$$

$$= -1.9739 \times 10^{-4}$$

$$= \int_{-0.02}^{0.02} \int_0^{2\pi} \int_0^{0.01} (-5 \times 10^{-6}) r dr d\theta dz$$

$$= (-5 \times 10^{-6}) \int_{-0.02}^{0.02} \int_0^{2\pi} \left(\frac{10^{-4}}{2} \right) d\theta dz$$

$$= \left(-\frac{5}{2} \right) (10^{-10}) \int_{-0.02}^{0.02} (2\pi) dz$$

$$= (-2.5) (10^{-10}) (2\pi) (0.02)$$

$$= -3.1416 \times 10^{-11} \text{ C}$$

$E(r) = ?$ source pt is located at $(0,1,0)$
find $E(r)$ at $(0,0,1)$

$$r = (0,1,0) \quad r' = (0,0,1)$$

$$\rho = \sqrt{x^2 + y^2} = 1 \quad \rho = \sqrt{x'^2 + y'^2} = 0$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) = 0 \quad \phi = \tan^{-1}\left(\frac{y'}{x'}\right) = 0$$

$$z = 0 \quad z = 1$$

$$(1,0,0) \quad (0,0,1)$$

$$(r-r') = (1,0,-1)$$

$$|r-r'| = \sqrt{2}$$

$$E(r) = \int_V \rho_v dv \frac{1}{4\pi\epsilon_0 |r-r'|^2} \frac{r-r'}{|r-r'|}$$

$$= (-3.1416 \times 10^{-11}) \left(\frac{1}{4\pi\epsilon_0 (\sqrt{2})^3} \right) \frac{(1,0,-1)}{\sqrt{2}}$$

$$= -0.0998 (1,0,-1)$$

$$= (-0.0998, 0, 0.998)$$

Line Charge

$$\rho_L = \frac{\text{Total charge}}{\text{Total length}} = \frac{dQ}{dL}$$

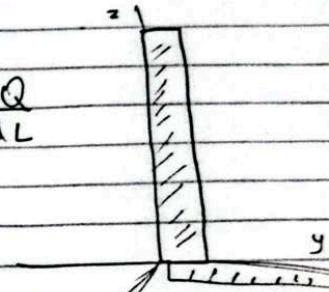
$$dQ = \rho_L dL$$

$$\Rightarrow Q = \int \rho_L dL$$

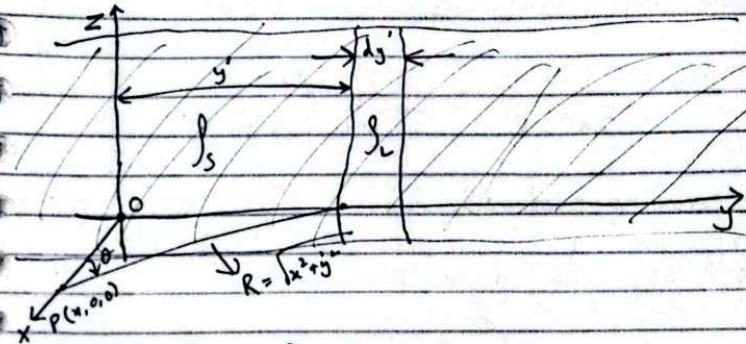
$$\Rightarrow Q = \int_x \rho_L dx = \int_y \rho_L dy = \int_z \rho_L dz$$

$$\Rightarrow Q = \int \rho_L dy = \int \rho_L dz$$

$$\Rightarrow Q = \int \rho_L dr$$



Field of a Sheet Charge



Surface Charge Density

Amount of electric charge per unit area of a surface (C/m^2)

Infinite Sheet of Charge

Lets place an infinite charge in the yz -plane with uniform charge density ρ_s .

⇒ Due to the symmetry, the electric field cannot vary along y -axis or z -axis because any point on the sheet looks identical along these directions.

The line charge density of the thin strip (in infinite sheet of charge)

$$\rho_l = \rho_s dy'$$

$$R = \sqrt{x^2 + y'^2}$$

Using Coulomb's law, the differential contribution to the electric field dE_x from the strip is

$$dE_x = \frac{\rho_s x}{2\pi\epsilon_0 R^2} = \frac{\rho_s dy' x}{2\pi\epsilon_0 (x^2 + y'^2)}$$

Integrating over all the differential strips along the y-axis gives the total field at E_x at point 'P'

$$E_x = \int_{-\infty}^{\infty} dE_x = \int_{-\infty}^{\infty} \frac{\rho_s dy' x}{2\pi\epsilon_0 (x^2 + y'^2)} = \frac{\rho_s x}{2\pi\epsilon_0} \int_{-\infty}^{\infty} \frac{dy'}{x^2 + y'^2}$$

$$= \frac{\rho_s x}{2\pi\epsilon_0} \left[\frac{1}{x} \tan^{-1}\left(\frac{y'}{x}\right) \right]_{-\infty}^{\infty}$$

$$= \frac{\rho_s x}{2\pi\epsilon_0} \left(\frac{\pi}{x} \right)$$

$$E_x = \frac{\rho_s}{2\epsilon_0} \hat{a}_x$$

Point Charge:

$$E = \frac{Q}{4\pi\epsilon_0 R^2} \hat{a}_R$$

Line Charge:

$$E = \frac{\rho_L}{2\pi\epsilon_0 R} \hat{a}_R$$

Sheet Charge:

$$E = \frac{\rho_s}{2\epsilon_0} \hat{a}_n$$

Sheet of Charge

Problem

Three uniform sheets of charge are located in free space as follows

- | | |
|----------------------------------|-----------------------------|
| ① 3 nC/m^2 at $z = -4$ | Find E at $P_A(2, 5, -1)$ |
| ② 6 nC/m^2 at $z = 1$ | at $P_E(-1, 5, 2)$ |
| ③ -8 nC/m^2 at $z = 4$ | at $P_B(4, -3)$ |
| | at $P_D(-2, 4, 5)$ |

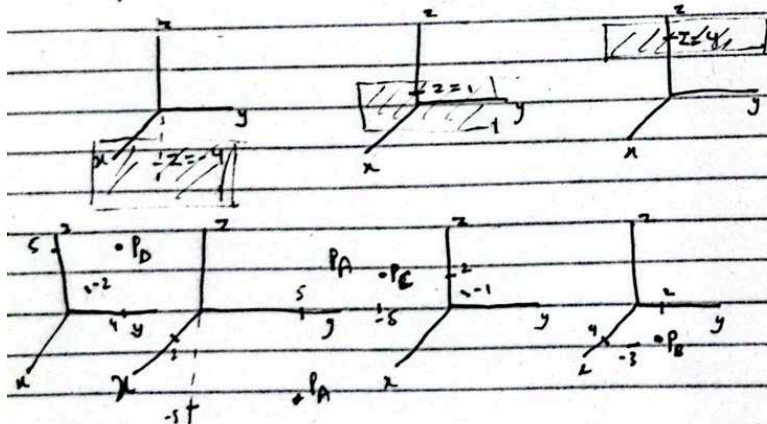
$$E = \frac{\rho_s}{2\epsilon_0} \cdot a_n$$

$$E_1 = \frac{3 \text{ n}}{2(8.854 \times 10^{-12})} = 169.414 \text{ V/m}$$

$$E_2 = \frac{6 \text{ n}}{2(8.854 \times 10^{-12})} = 338.829 \text{ V/m}$$

$$E_3 = \frac{-8 \text{ n}}{2(8.854 \times 10^{-12})} = -451.97 \text{ V/m}$$

$$E_r = 169.414 + 338.829 - 451.97 = 56.273 \text{ V/m}$$



$$E_{P_A} = -E_1 - E_2 + E_3$$

$$E_{P_B} = -E_1 - E_2 + E_3$$

$$E_{P_C} = -E_1 - E_2 + E_3 = E_{P_D}$$

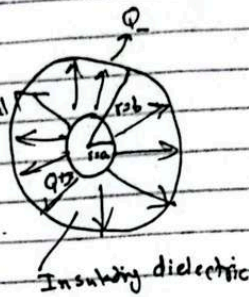
CN#3

Electric flux Density

Electric flux (Ψ) is directly proportional to charge on the inner space.

Unit: Coulomb

(Displacement flux) $\Psi = Q$



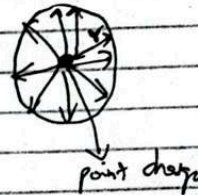
→ Electric flux density (D) is defined as the electric flux per unit area (C/m^2).

→ Electric flux at any given point is given by the no. of flux lines crossing a surface normal to the lines divided by the surface.

→ Due to point charge, electric flux density is

$$D = \frac{\Psi}{4\pi r^2} \hat{a}_r$$

$$= \frac{Q}{4\pi r^2} \hat{a}_r$$



$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2}; D = \frac{Q}{4\pi\epsilon_0 r^2} \hat{a}_r \quad E.$$

$$D = E\epsilon_0 \quad \text{due to point charge for free space}$$

$$D = E\epsilon_0\epsilon_r \quad \text{for air permittivity of air}$$

(Volume charge density)

$$E = \int \frac{\rho_v dv}{4\pi\epsilon_0 r^2} \hat{a}_r$$

$$EE_0 = \int \frac{\rho_v dv}{4\pi\epsilon_0 r^2} \hat{a}_r$$

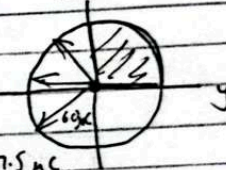
$$\Rightarrow D = \int \frac{\rho_v dv}{4\pi r^2} \hat{a}_r$$



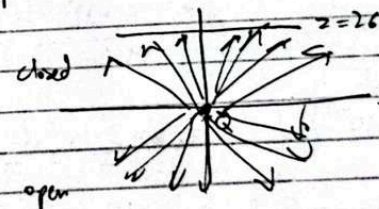
- Q) Given a $60\mu C$ point charge located at the origin, find the total electric flux passing through flux passing through
- a) position of sphere $r = 26cm$ bounded by $0 < \theta < \frac{\pi}{2}$ and $0 < \phi < \frac{\pi}{2}$

Sol $\frac{1}{8}$ th section of total sphere

$$\Psi = Q \Rightarrow \Psi = Q \times \frac{1}{8} = 60 \times \frac{1}{8} = 7.5 \mu C$$



c) plane $z = 26cm$



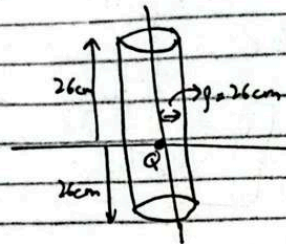
$$\Psi = Q$$

$$\Rightarrow \Psi = \frac{1}{2} Q$$

$$\Psi = 30 \mu C$$

(b) Closed surface defined $y = 26 \text{ cm}$, $z = \pm 26 \text{ cm}$

$$\Psi = Q = 60 \mu\text{C}$$



Q) Calculate $\vec{D}$ in RCS at point $P(2, -3, 6)$

a) Due to point charge, $Q_A = 55 \text{ nC}$ at $Q(-2, 3, 6)$

$$\vec{D} = \frac{Q}{4\pi r^2} \hat{r}$$

$$\vec{r} = \vec{AP} = (2 - (-2), -3 - 3, 6 - 6) = (4, -6, 0)$$

$$|\vec{r}| = \sqrt{(4)^2 + (-6)^2 + (0)^2} = 7.21 \text{ m}$$

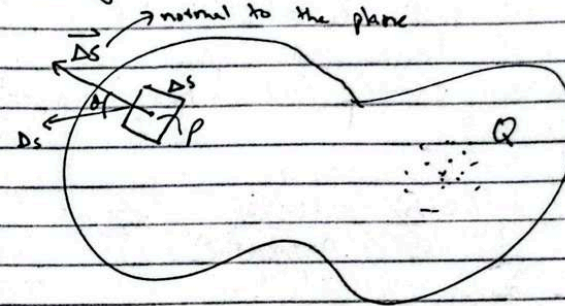
$$\vec{D} = \frac{55 \text{ nC}}{4\pi (7.21)^2} \frac{(4, -6, 0)}{7.21}$$

$$= (2.23 \times 10^{-5}) \left(\frac{4}{7.21} \hat{a}_x - \frac{6}{7.21} \hat{a}_y + \frac{0}{7.21} \hat{a}_z \right)$$

$$= 6.37 \hat{a}_x - 9.56 \hat{a}_y + 0 \hat{a}_z \text{ nC/m}^2$$

Gauss's Law

"The electric flux passing through any closed surface is equal to the total charge enclosed by that surface."



$$\Delta \Psi = \text{flux passing through } \Delta S \\ = E \Delta S \cos \theta = E \cdot \Delta S$$

$$\text{Total flux} = \Psi = \int d\Psi = \oint_{\text{closed surface}} E \cdot \Delta S = \oint_{\text{closed surface}} E \cdot dS$$

closed surface
integral

RC $\Psi = Q = \iint_R D_{xy} dx dy$

CS $\Psi = Q = \iint_S D_{\theta, \phi} d\theta d\phi$

SCS $\Psi = Q = \int_0^\pi \int_0^{2\pi} D_{\theta, \phi} r^2 \sin \theta d\theta d\phi$

Point Charge $\Psi = Q = q_1 + q_2 + q_3 + \dots + q_n \\ = \sum Q_n = \oint D_s \cdot dS$

Line Charge $\Psi = Q = \int_L dl = \oint D_s \cdot dS$

Surface Charge $\Psi = Q = \int_S dS = \oint D_s \cdot dS$

Volume charge distribution

$$\varphi = Q = \int_V \rho_v dv = \oint_S D_s \cdot dS$$

D3.3

Q) Given $D = 0.3r^2 \hat{a}_r$ nC/m² in free space

- Find E at point $P(r=2, \theta=25^\circ, \phi=90^\circ)$
- Find total charge within sphere $r=3$
- Find total electric flux leaving sphere $r=4$

$$D = \epsilon E$$

$$E = \frac{D}{\epsilon} = \frac{0.3r^2}{8.85 \times 10^{-12}} \hat{a}_r = 0.305 \hat{a}_r$$

(a) $D = \frac{Q}{4\pi r^2} \hat{a}_r$

$$Q = 4\pi r^2 D \hat{a}_r$$

$$= 4\pi (2)^2 (0.3) (10^{-9})$$

$$= 305 \times 10^{-9} C$$

$$Q = 305 nC$$

(a) $D = \epsilon E$

$$E = \frac{D}{\epsilon} = \frac{0.3(2)^2 (10^{-9})}{8.85 \times 10^{-12}} = 135.532 \hat{a}_r V/m$$

(b) $D = \frac{Q}{4\pi r^2} \hat{a}_r$

$$4\pi r^2 \times 0.3 \hat{a}_r = Q$$

$$Q = 4\pi (3)^2 (0.3) (10^{-9})$$

$$= 305 \times 10^{-9} C$$

$$Q = 305 nC$$

(c) $D = \frac{Q}{4\pi r^2} \hat{a}_r$

$$= \frac{305 nC}{4\pi (4)^2} \hat{a}_r$$

$$= 0.305 \hat{a}_r$$

Eg 3.1

$$(2\pi)(2\pi) \quad 2D_2(2\pi) = 829$$

$$Q = (829) \epsilon_0$$

$$= \frac{829}{4\pi\epsilon_0}$$

(c) $\varphi = \frac{Q}{4\pi\epsilon_0 r}$

$$D = \frac{Q}{4\pi r^2} \hat{a}_r$$

$$Q = (0.3)(4)^2 (4\pi)(4)^2 (10^{-9})$$

$$= 965 nC$$

Eg 7.1) $D = \frac{Q}{4\pi r^2} \hat{a}_r$

$$\Rightarrow D_s = \frac{Q}{4\pi a^2} \hat{a}_r$$

$$\Rightarrow dS = r^2 \sin\theta d\theta d\phi = a^2 \sin\theta d\theta d\phi$$

$$\text{or } dS = a^2 \sin\theta d\theta d\phi \hat{a}_r$$

$$D_s \cdot dS = \frac{Q}{4\pi a^2} a^2 \sin\theta d\theta d\phi \hat{a}_r \cdot \hat{a}_r$$

$$= \frac{Q}{4\pi} \sin\theta d\theta d\phi$$

$$\int_0^{2\pi} \int_0^\pi \frac{Q}{4\pi} \sin\theta d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^\pi \frac{Q}{4\pi} \sin\theta d\theta d\phi$$

$$= \int_0^{2\pi} \frac{Q}{4\pi} (-\cos\theta)_0^\pi d\phi = \int_0^{2\pi} \frac{Q}{2\pi} d\phi = Q$$

34-35

Applications of Gauss's Law: Symmetrical Charge Distributions

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q \rightarrow \text{only valid/applicable for symmetrical surfaces}$$

Symmetry (spherical, cylindrical, planar symmetry)

Key Conditions for Applying Gauss's Law

To effectively use Gauss's law, we must choose a Gaussian surface that satisfies the following conditions:

- ① Electric flux density must be either normal or tangential to the surface

If $\mathbf{D}$ is normal to surface,
 $\mathbf{D} \cdot d\mathbf{S} = D dS$

If $\mathbf{D}$ is tangential to surface,
 $\mathbf{D} \cdot d\mathbf{S} = 0$

- ② The magnitude of $\mathbf{D}$ must be constant over the surface where it is normal.

→ This allows us to pull D outside of the integral, leaving only the surface area inside the integral.

①

$$Q = \oint \mathbf{D} \cdot d\mathbf{S} = \oint$$

$$= D_s \int dS$$

$$= D_s \int_{\text{top}} dS + D_s \int_{\text{bottom}} dS + D_s \int_{\text{side}} dS$$

$$= 0 \int_{\text{top}} dS + 0 \int_{\text{bottom}} dS + D_s \int_{\text{side}} dS = D_s \int_{\phi=0}^{2\pi} \int_{z=0}^L \rho d\phi dz$$

$$= D_s 2\pi \rho L$$

$$Q = D_s 2\pi \rho L$$

$$D_s = \frac{Q}{2\pi \rho L}$$

$$D_s = D_p = \frac{Q}{2\pi \rho L}$$

Also for Line charge

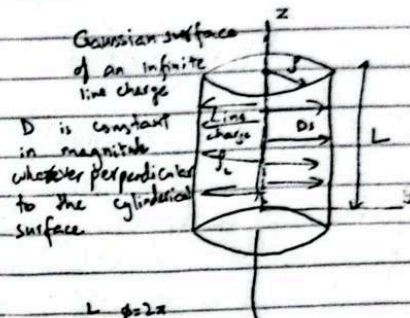
$$Q = \int_L \lambda dz$$

$$\text{then } D_s = D_p = \frac{\lambda \cdot L}{2\pi \rho L} = \frac{\lambda}{2\pi \rho}$$

$$D_p = \frac{\lambda}{2\pi \rho}$$

As $D = \epsilon E$

$$\Rightarrow E = \frac{\lambda}{2\pi \epsilon_0 \rho}$$

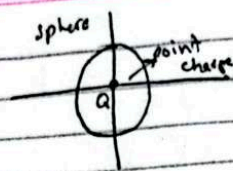


$$\textcircled{2} \quad Q = \oint \mathbf{D}_s \cdot d\mathbf{s} = \oint D_s ds$$

$$= D_s \oint ds = D_s 4\pi r^2$$

$$D_s = \frac{Q}{4\pi r^2} \hat{\mathbf{a}}_r$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \hat{\mathbf{a}}_r$$



③ Coaxial Cable with Cylindrical Symmetry

$$D_s = D_p = \frac{Q}{2\pi a l}$$

$$\Rightarrow Q = D_s 2\pi a l$$

The total charge on a length L of the inner conductor is

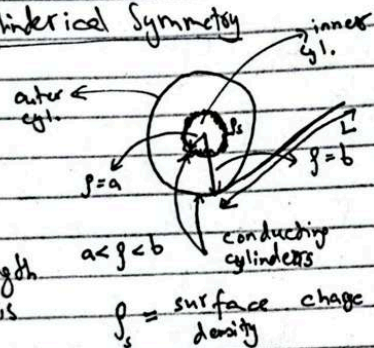
$$Q = \int_{z=0}^L \int_{\phi=0}^{2\pi} \rho_s a d\phi dz$$

$$= 2\pi a l \rho_s = Q$$

$$\Rightarrow D_s 2\pi a l = 2\pi a l \rho_s$$

$$D_s = \frac{a \rho_s}{l}$$

$$D_s = \frac{a \rho_s}{l} \hat{\mathbf{a}}_r$$



.. Eg 3.2, 3.1

$$Q_{\text{enc inner}} = 2\pi a l \rho_s$$

$$Q_{\text{enc outer}} = -Q_{\text{enc inner}} = -2\pi a l \rho_s$$

$$2\pi b l \rho_s = -2\pi a l \rho_s$$

$$\frac{\rho_{s \text{ outer}}}{\rho_{s \text{ inner}}} = \frac{-a}{b}$$

16-10 14-15 133 128
130 2

Application of Gauss's Law: Differential Volume Element

for symmetrical surfaces, electric field or electric flux is constant (Gaussian Surface).

What if the surface is not symmetrical?

We will consider a very small differential volume element so that Gauss's Law can be applied on it to determine electric flux.

Gauss's Law:

$$\oint \mathbf{D} \cdot d\mathbf{s} = \int_{\text{front}} \mathbf{D} \cdot d\mathbf{s} + \int_{\text{back}} \mathbf{D} \cdot d\mathbf{s} + \int_{\text{Right}} \mathbf{D} \cdot d\mathbf{s} + \int_{\text{left}} \mathbf{D} \cdot d\mathbf{s}$$

$$+ \int_{\text{Top}} \mathbf{D} \cdot d\mathbf{s} + \int_{\text{bottom}} \mathbf{D} \cdot d\mathbf{s}$$

$$S = L \times B$$

$$\int_{\text{front}} \mathbf{D} \cdot d\mathbf{s} = \mathbf{D}_{\text{front}} \cdot \Delta \mathbf{s}_{\text{front}} = \mathbf{D}_{\text{front}} \cdot \Delta y \Delta z \mathbf{a}_x$$

$$\Delta \mathbf{s}_{\text{front}} = \Delta z \times \Delta y$$

$$\Delta \mathbf{s}_{\text{bottom}} = \Delta y \times \Delta x$$

Now to find $\mathbf{D}_{\text{front}}$ (D_x at the front surface). The front face is at a distance of $\frac{\Delta x}{2}$ from point P.

$$\mathbf{D}_{\text{front}} = \mathbf{D}_x + \frac{\Delta x}{2} \times \text{Rate of change } D_x \text{ with } x$$

$$= \mathbf{D}_x + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}$$

where $\mathbf{D}_x$ is value of D_x at P

and Partial derivative is used to express the rate of change of D_x with x .

(Reaction) 1 2-a 107 100
0 open
1 full
2 own

$$\mathbf{D}_{\text{front}} = \left(\mathbf{D}_x + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

Now to find $\mathbf{D}_{\text{back}}$ (D_x at back of surface). The back face is at a distance of $-\frac{\Delta x}{2}$ from Point P

$$\int_{\text{back}} \mathbf{D} \cdot d\mathbf{s} = \mathbf{D}_{\text{back}} \cdot \Delta \mathbf{s}_{\text{back}} = \mathbf{D}_{\text{back}} \cdot (\Delta y \Delta z (-\mathbf{a}_x))$$

$$= -\mathbf{D}_{\text{back}} \Delta y \Delta z$$

$$\mathbf{D}_{\text{back}} = \mathbf{D}_x - \frac{\Delta x}{2} \frac{\partial D_x}{\partial x}$$

$$\int_{\text{back}} \mathbf{D} \cdot d\mathbf{s} = \left(-\mathbf{D}_x + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

Now along x-axis combine both eq.

$$\mathbf{D}_{\text{front}} + \mathbf{D}_{\text{back}} = \left(\mathbf{D}_x + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z + \left(-\mathbf{D}_x + \frac{\Delta x}{2} \frac{\partial D_x}{\partial x} \right) \Delta y \Delta z$$

$$\text{Along x-axis} \Rightarrow \frac{\partial D_x}{\partial x} \Delta x \Delta y \Delta z$$

Along y-axis:

$$\oint_{\text{y-axis}} \mathbf{D} \cdot d\mathbf{s} = \frac{\partial D_y}{\partial y} \Delta x \Delta y \Delta z$$

Along z-axis

$$\oint_{\text{z-axis}} \mathbf{D} \cdot d\mathbf{s} = \frac{\partial D_z}{\partial z} \Delta x \Delta y \Delta z$$

$$\oint_{\text{total}} \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enclosed}}$$

$$Q_{\text{enclosed}} = \underbrace{\left(\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} \right)}_{\text{divergence}} \Delta x \Delta y \Delta z$$

$$Q_{\text{enc}} = \nabla \cdot \mathbf{D} \cdot \Delta V$$

$$\frac{Q_{\text{enc}}}{\Delta V} = \nabla \cdot \mathbf{D}$$

$$\nabla \cdot \mathbf{D} = \rho_v$$

Divergence Theorem

The Del operator (∇) is a vector operator used to perform several important operations in vector calculus, including divergence, gradient and curl.

$$\nabla = \frac{\partial}{\partial x} \hat{a}_x + \frac{\partial}{\partial y} \hat{a}_y + \frac{\partial}{\partial z} \hat{a}_z$$

→ $\hat{a}_x$, $\hat{a}_y$ and $\hat{a}_z$ are unit vectors in the x, y, z directions.

→ Partial derivatives indicate rates of change in the x respective direction.

The del operator behaves like a vector but instead of returning scalar, it returns partial derivatives when applied to scalar and vector fields.

- 1) Gradient (∇u) → applied to scalar field "u" → yielding a vector
- 2) Divergence ($\nabla \cdot \mathbf{D}$) → applied to vector field → yielding a scalar

Divergence theorem connects the behavior of a vector field inside a volume to the behavior of the surface enclosing that volume. Mathematically,

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{V_01} (\nabla \cdot \mathbf{D}) dV$$

double integral

• $\oint_S \mathbf{D} \cdot d\mathbf{S}$ represents the flux through the surface S enclosing volume

triple integral

• $\int_{V_01} (\nabla \cdot \mathbf{D}) dV$ is the integral of divergence $\mathbf{D}$ throughout the volume

The divergence theorem states that the total flux exiting a close surface is equal to the integral of divergence of field within the volume enclosed by that surface.

Divergence:-

In different coordinate system
In cylindrical coordinate system

$$\nabla \cdot D = \frac{1}{r} \frac{\partial}{\partial r} (r D_r) + \frac{1}{r} \frac{\partial D_\theta}{\partial \theta} + \frac{\partial D_z}{\partial z}$$

Gauss's Law:-

$$\oint_S D \cdot dS = Q_{enc} = \int_{vol} (\nabla \cdot D) dV$$

$$\oint_S D \cdot dS = \int_{vol} (\nabla \cdot D) dV$$

Divergence and Maxwell's Equation:-

$$\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \lim_{\Delta V \rightarrow 0} \oint \frac{D \cdot dS}{\Delta V} = \lim_{\Delta V \rightarrow 0} \frac{Q_{enc}}{\Delta V}$$

$$\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = \lim_{\Delta V \rightarrow 0} \oint \frac{A \cdot dS}{\Delta V} \rightarrow A \text{ could be velocity force or any other vector quantity}$$

Divergence of A = $\text{div } A = \lim_{\Delta V \rightarrow 0} \oint \frac{A \cdot dS}{\Delta V}$ Maxwell's Eq.

SCS $\rightarrow$

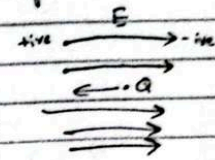
$$\text{div } D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta D_\theta) + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

Ch#4

4.1 Energy Expended in moving a charge in an Electric field

Force exerted by the field $F_E = QE \rightarrow \text{+ve to -ve}$

Force to move charge against field: $F_{app} = -QE$



Work done in Moving a Charge

Moving the charge Q in a small distance dL within an electric field E, the component of the force in the direction of dL is

$$F_L = F_E \cdot \hat{L} \quad (F_E \cdot \hat{a})$$

$\hat{L}$ is a unit vector in the direction of dL

$$F_{app} = -QE \cdot \hat{L}$$

The differential 'dW' done by the applied in moving the charge by a small distance dL is given by

$$dW = F_{app} \cdot dL$$

$$dW = -QE \cdot dL$$

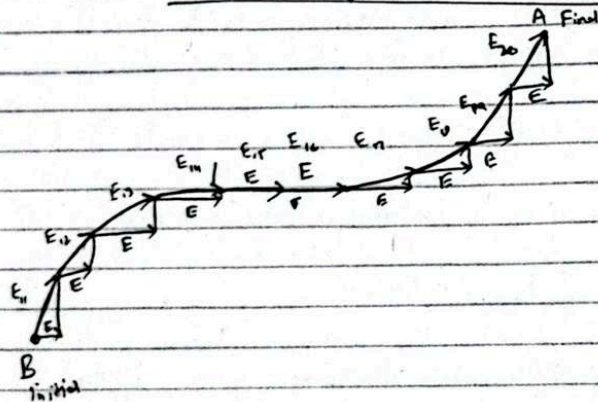
Condition when the work done is zero:-

- When E and dL are perpendicular
- When Q = 0
- If dL = 0
- If E = 0

$$W = -Q \int_{\text{Initial}}^{\text{Final}} \mathbf{E} \cdot d\mathbf{L}$$

Work done over a finite distance

4.2 Line Integral In a Uniform Electric field



Assuming uniform electric field consider the path b/w A and B we want to calculate the work done in moving a charge point Q along its path.

$$W = -Q \int_B^A \mathbf{E} \cdot d\mathbf{L}$$

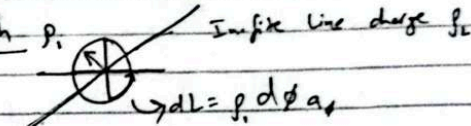
$$W = -QE \int_B^A dL \Rightarrow \int_B^A dL = L_{BA}$$

$$W = -QE \cdot L_{BA}$$

This means the work done in a uniform electric field depends only on L_{BA} (path from A to B).

In Circular Path

①



$$W = -Q \int_0^{2\pi} \mathbf{E} \cdot d\mathbf{L}$$

$$\mathbf{E} = E_r \hat{a}_r = \frac{\lambda}{2\pi\epsilon_0 r} \hat{a}_r$$

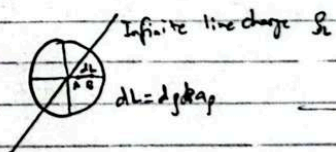
$$d\mathbf{L} = r d\phi \hat{a}_\phi$$

$$W = -Q \int_0^{2\pi} \frac{\lambda}{2\pi\epsilon_0 r} \hat{a}_r \cdot r d\phi \hat{a}_\phi$$

$$= -Q \int_0^{2\pi} \frac{\lambda}{2\pi\epsilon_0} d\phi \hat{a}_r \cdot \hat{a}_\phi \quad \because \hat{a}_r \cdot \hat{a}_\phi = 0$$

$W = 0$ in circular path

②



$$\mathbf{E} = E_r \hat{a}_r = \frac{\lambda}{2\pi\epsilon_0 r} \hat{a}_r$$

$$d\mathbf{L} = r d\phi \hat{a}_\phi$$

$$\begin{aligned}
 W &= -Q \int_A^B \mathbf{E} \cdot d\mathbf{L} \\
 &= -Q \int_A^B \frac{Q}{2\pi\epsilon_0 s} \hat{\mathbf{a}}_s \cdot d\mathbf{L} \\
 &= -Q \int_A^B \frac{Q}{2\pi\epsilon_0} \frac{ds}{s} \hat{\mathbf{a}}_s \cdot \hat{\mathbf{a}}_s \quad \because \hat{\mathbf{a}}_s \cdot \hat{\mathbf{a}}_s = 1 \\
 &= -Q \int_A^B \frac{Q}{2\pi\epsilon_0} \frac{ds}{s} \\
 &= -\frac{Q^2}{2\pi\epsilon_0} \int_A^B \frac{ds}{s}
 \end{aligned}$$

$$W = -\frac{Q^2}{2\pi\epsilon_0} \ln \frac{B}{A}$$

Line Integral in different Coordinate Systems:-

RCS: $d\mathbf{L} = dx \hat{\mathbf{x}} + dy \hat{\mathbf{y}} + dz \hat{\mathbf{z}}$

CCS: $d\mathbf{L} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin\theta d\phi \hat{\boldsymbol{\phi}}$

SCS: $d\mathbf{L} = dr \hat{\mathbf{r}} + r d\theta \hat{\boldsymbol{\theta}} + r \sin\theta d\phi \hat{\boldsymbol{\phi}}$

4.3

Potential Difference and Potential

Potential difference is the work done by an external force to move a unit +ve charge from one point to another in an electric field.

$$\begin{aligned}
 W &= -Q \int \mathbf{E} \cdot d\mathbf{L} \\
 V &= - \int_{\text{Initial}}^{\text{final}} \mathbf{E} \cdot d\mathbf{L} \\
 V &= - \int_A^B \mathbf{E} \cdot d\mathbf{L} \\
 V &= \frac{W}{Q}
 \end{aligned}$$

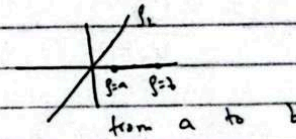

Unit: Volts

Example

$$\mathbf{E} = \frac{Q}{2\pi\epsilon_0 s} \hat{\mathbf{a}}_s$$

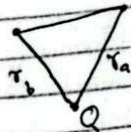
$$W = \frac{Q^2}{2\pi\epsilon_0} \ln \left(\frac{b}{a} \right)$$

$$\frac{W}{Q} = V_{ab} = \frac{Q}{2\pi\epsilon_0} \ln \left(\frac{b}{a} \right)$$



Example

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \hat{a}_r$$



$$dL = dr \hat{a}_r$$

$$V_{AB} = - \int_{r_b}^{r_a} \frac{Q}{4\pi\epsilon_0 r^2} dr$$

$$= - \frac{Q}{4\pi\epsilon_0} \int_{r_b}^{r_a} \frac{1}{r^2} dr$$

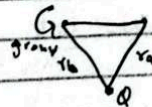
$$= - \frac{Q}{4\pi\epsilon_0} \left(- \frac{1}{r_a - r_b} \right)$$

potential) $V_{AB} = \frac{Q}{4\pi\epsilon_0} \frac{1}{r_a - r_b}$

Absolute Potential (w.r.t ground)

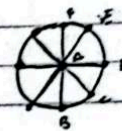
$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_a} \right)$$

$$\Rightarrow \boxed{V = \frac{Q}{4\pi\epsilon_0 r}}$$



Equipotential Surfaces

- Consists of all the points in space that have the same value of potential
- No work is required to move a charge along an equipotential surface because the potential difference b/w any two points on the surface is zero.



$$\Rightarrow V = \frac{Q}{4\pi\epsilon_0 r} \quad \begin{matrix} \nearrow \text{energy} \\ \Rightarrow V \propto \frac{1}{r} \end{matrix}$$

$$\Rightarrow E = \frac{Q}{4\pi\epsilon_0 r^2} \quad \begin{matrix} \nearrow \text{force} \\ \Rightarrow E \propto \frac{1}{r^2} \end{matrix}$$

$$F = qE$$

Potential Diff. of a System of Charges and Conservative Property

Potential due to a single point Q, located at position r_1

$$V(r) = \frac{Q_1}{4\pi\epsilon_0 |r - r_1|}$$

$V(r)$ is potential at distance r from charge Q_1

Superposition of Potential Fields:-

For single point charge

$$V(r) = \frac{Q_1}{4\pi\epsilon_0 |r-r_1|}$$

For system of multiple charges

→ For two charges

$$V(r) = \frac{Q_1}{4\pi\epsilon_0 |r-r_1|} + \frac{Q_2}{4\pi\epsilon_0 |r-r_2|}$$

→ For n charges

$$V(r) = \sum_{n=1}^m \frac{Q_n}{4\pi\epsilon_0 |r-r_n|}$$

4.6 Potential Gradient

General Line Integral Relationship

$$V = - \int E \cdot dL$$

$$\Delta V = -E \cdot \Delta L = -E \Delta L \cos \theta$$

When ΔV is maximized,

- When $\cos \theta = -1$, $\theta = 180^\circ \rightarrow$ mean ΔL points in the direction opposite to E .

- Magnitude of E at any point is given by the maximum rate of change of potential w.r.t distance. $\left(\frac{dV}{dL}\right)_{\max}$

Equipotential Surface

$$\Delta V = E \cdot \Delta L = 0$$

$$= E \Delta L \cos \theta = 0$$

$$\Rightarrow \theta = 90^\circ$$

E is perpendicular to ΔL in equipotential surface and E points in direction where potential decreases.

$$\Delta V = -E \cdot \Delta L$$

$$E = - \frac{\Delta V}{\Delta L} = - \frac{dV}{dL}$$

where $\frac{dV}{dL}$ is the rate of change of V w.r.t L along normal

$$E = \left(\frac{dV}{dL}\right)_{\max} \hat{a}_N = - \frac{dV}{dN} \hat{a}_N$$

where $\frac{dV}{dL}$ represents max. rate of change of V w.r.t L along normal direction

Gradient of Scalar Field

The operation of finding E from V is an example of gradient operation which is a vector operation applied to a scalar field.

Generic form:

$$\text{grad } T = \nabla T = \frac{dT}{dN} \mathbf{a}_N$$

for relationship b/w E and V

$$E = -\nabla V$$

In different coordinate system,

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

$$E = E_x \mathbf{a}_x + E_y \mathbf{a}_y + E_z \mathbf{a}_z$$

$$E_x = -\frac{\partial V}{\partial x} \quad E_y = -\frac{\partial V}{\partial y} \quad E_z = -\frac{\partial V}{\partial z}$$

RCS

$$E = -\nabla V = -\left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z\right)$$

CCS

$$E = -\nabla V = -\left(\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z\right)$$

CS

$$E = -\nabla V = -\left(\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi\right)$$

Q) Given $V = 2x^2y - 5z$

Point $P(-4, 3, 6)$

At point P , what is V, E , direction of E , D & $\oint_V$?

$$\Rightarrow V = 2(-4)^2(3) - 5(6) = 66 \text{ W}$$

$$E = -\nabla V = -\left(\frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z\right)$$

$$= -\left(4xy \mathbf{a}_x + 2x^2 \mathbf{a}_y - 5 \mathbf{a}_z\right)$$

$$= -4xy \mathbf{a}_x - 2x^2 \mathbf{a}_y + 5 \mathbf{a}_z$$

$$= -48 \mathbf{a}_x - 32 \mathbf{a}_y + 5 \mathbf{a}_z$$

$$|E| = 57.9 \text{ V/m}$$

$$\hat{E} = \frac{\vec{E}}{|E|} = \frac{-48 \mathbf{a}_x - 32 \mathbf{a}_y + 5 \mathbf{a}_z}{57.9} \quad (\text{direction of } E)$$

$$D = E \epsilon_0 = (-48 \mathbf{a}_x - 32 \mathbf{a}_y + 5 \mathbf{a}_z) (8.854 \times 10^{-12})$$

$$\oint_V = \nabla \cdot D = -35.4 \text{ nC} - 17.7 \text{ nC} + 44.3 \text{ nC} = (-35.4 \text{ nC})$$

4.7

Electric Dipole

The total potential at point P is sum of potentials due to +Q and -Q.

- Distance from +Q to P: R_1

- Distance from -Q to P: R_2

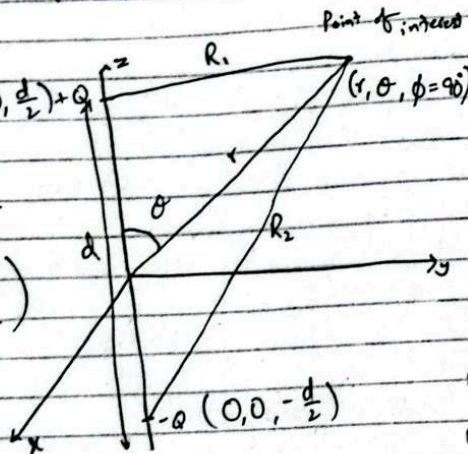
Using superposition,

$$V = V_{+Q} + V_{-Q} \quad (0, 0, \frac{d}{2}) + Q \quad (r, \theta, \phi = 90^\circ)$$

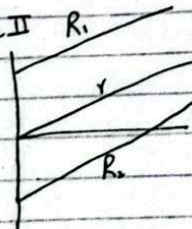
$$= \frac{Q}{4\pi\epsilon_0 R_1} + \frac{(-Q)}{4\pi\epsilon_0 R_2}$$

$$= \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$= \frac{Q(R_2 - R_1)}{4\pi\epsilon_0 R_1 R_2}$$



Case II



If point is very distant

$$R_1 \approx R_2 \approx r$$

$$V = \frac{Q(R_2 - R_1)}{4\pi\epsilon_0 R_1 R_2}$$

$$V = \frac{Q d \cos \theta}{4\pi\epsilon_0 r^2}$$

$R_2 - R_1 = d \cos \theta$
(projection of R_1 on R_2 on x-axis)

$$\text{At } E = -\nabla V = - \left(\frac{\partial V}{\partial x} \hat{a}_x + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{a}_\phi \right)$$

$$\frac{\partial V}{\partial r} = \frac{\partial}{\partial r} \left(\frac{Q d \cos \theta}{4\pi\epsilon_0 r^2} \right)$$

$$= \frac{-2 Q d \cos \theta}{4\pi\epsilon_0 r^3}$$

$$= - \frac{Q d \cos \theta}{2\pi\epsilon_0 r^3}$$

$$\frac{\partial V}{\partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{Q d \cos \theta}{4\pi\epsilon_0 r^2} \right)$$

$$= \frac{-Q d \sin \theta}{4\pi\epsilon_0 r^2}$$

$$\frac{\partial V}{\partial \phi} = \frac{\partial}{\partial \phi} \left(\frac{Q d \cos \theta}{4\pi\epsilon_0 r^2} \right)$$

$$= 0$$

$$E = - \left(\frac{-Q d \cos \theta}{2\pi\epsilon_0 r^3} \hat{a}_r - \frac{1}{r} \frac{Q d \sin \theta}{4\pi\epsilon_0 r^2} \hat{a}_\theta + \frac{1}{r \sin \theta} (0) \right)$$

$$E = \frac{Q d}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{a}_r + \sin \theta \hat{a}_\theta)$$

Dipole Moment $\vec{p}$

$$\vec{p} = Q \vec{d}$$

where $\vec{d}$ is the vector from $-Q$ to $+Q$

- Units of $\vec{p}$: Cm (coulomb metre)

$$- V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2} \hat{a}_r$$

$$- V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$\text{As } \theta = 0^\circ, \cos \theta = 1$$

$$r \approx |\vec{r} - \vec{r}'|$$

$$- V = \frac{p}{4\pi\epsilon_0 r^2} \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$V = \frac{\vec{p} \cdot (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} \rightarrow \frac{(\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

From origin, $\vec{r}' = 0$

$$V \propto \frac{1}{r^2}$$

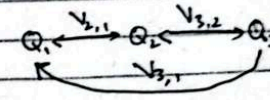
$$E \propto \frac{1}{r^3}$$

4.8 Energy Density in Electrostatic field

- Work done by an external source $\rightarrow$ Potential energy

- For doing work on multiple charges, $+Q \xrightarrow{d} +Q$

$$W_E = Q_1 V_{1,1} + Q_2 (V_{2,1} + V_{2,2}) + \dots$$



Expression for total P.E

$$W_E = \frac{1}{2} \sum_{m=1}^N Q_m V_m$$

Total potential energy W_E of N point charges

Potential energy for continuous charge distribution:-

$$W_E = \frac{1}{2} \int_V \rho_v V dv$$

Using Maxwell's Eq. $\nabla \cdot \vec{D} = \rho_v$

$$W_E = \frac{1}{2} \int_V (\nabla \cdot \vec{D}) V dv$$

Applying a vector identity

$$\nabla \cdot (DV) = V(\nabla \cdot D) + D \cdot (\nabla V)$$

$$V(\nabla \cdot D) = \nabla \cdot (DV) - D \cdot (\nabla V)$$

Hence

$$W = \frac{1}{2} \int_V [\nabla \cdot (DV) - D \cdot (\nabla V)] dv$$

$$\int_V (\nabla \cdot D) = \oint_S D \cdot dS$$

Using Gauss's Divergence theorem,

$$W_E = \frac{1}{2} \oint_{\text{surface}} (VD) \cdot dS = \frac{1}{2} \int_{\text{volume}} D \cdot (\nabla V) dV$$

Assuming the system is isolated and V tends to 0 at infinity, the surface integral becomes

$$W_E = 0 - \frac{1}{2} \int_V D \cdot (\nabla V) dV$$

$$W_E = -\frac{1}{2} \int_V D \cdot (\nabla V) dV$$

$$W_E = \frac{1}{2} \int_V D \cdot (-\nabla V) dV$$

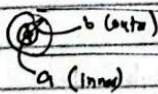
As $E = -\nabla V$

Energy in
electrostatic
field

$$W_E = \frac{1}{2} \int_V D \cdot E dV$$

① Energy stored in Co-axial Section

$$W_E = \pi L \frac{a^2 \rho_s^2}{\epsilon_0} \ln \frac{b}{a}$$



② Energy stored in capacitor

$$W_E = \frac{1}{2} QV$$

$$= \frac{1}{2} CV^2$$

As $Q = CV$

Ch#5

S.1

Current & Current Density

Current:-

Rate of flow of charge $I = \frac{dQ}{dt}$ [Unit 1A ($\frac{C}{s}$)]

Direction: +ve to -ve
⇒ Vector

Current Density:-

Rate of charge flow through a region (Volume)

⇒ Vector quantity ⇒ $J = \rho_v \vec{v}$

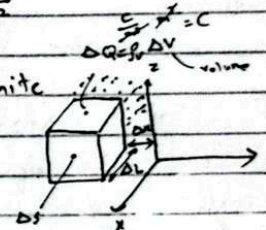
velocity of charge carrier
 $\frac{C}{m^2} = \frac{\frac{C}{m^3} \cdot m}{s} = \frac{C}{m^2 \cdot s} = \frac{A}{m^2}$

Relationship with Current

The current crossing an infinite surface dS normal to J is

$$dI = J \cdot dS$$

$$I = \int_S J \cdot dS$$



Current Density & Velocity Relationship:-

Consider a volume element of charge

$$\Delta Q = \rho_v \Delta V = \rho_v \Delta S \Delta L = \rho_v \Delta S \Delta x$$

Assuming the charge moves with velocity ' v ' through an incremental distance Δx in time Δt

$$\Delta I = \frac{\Delta Q}{\Delta t} = \frac{\rho_v \Delta S \Delta L}{\Delta t} = \rho_v \Delta S \frac{\Delta x}{\Delta t}$$

As $\frac{\Delta x}{\Delta t} = v_x \Rightarrow \Delta I = \rho_v \Delta S v_x$

For infinite conditions,

$$I = \rho_v v S$$

$$\Rightarrow I = \oint_S \mathbf{J} \cdot d\mathbf{S}$$

$$\frac{I}{S} = \rho_v \cdot v$$

$$\Rightarrow \boxed{J = \rho_v v} \quad \text{velocity}$$

S.2

Current Continuity

Integral form of Continuity Equation

$$I = \int_V \mathbf{J} \cdot d\mathbf{S}$$

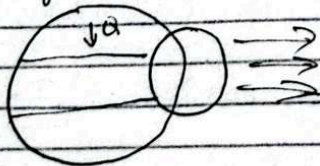
where

$\mathbf{J}$: current density vector ($\frac{A}{m^2}$)

From the conservation of charge principle, the rate of charge leaving the region must equal the decrease in the charge within the volume, i.e. Q represented the charge inside the closed surface.

$$\int_S \mathbf{J} \cdot d\mathbf{S} = -\frac{dQ_S}{dt}$$

The negative sign indicates the outward current corresponds to a decrease in the charge within the enclosed region.



Differential form of Continuity Equation

Divergence Theorem

$$\int_V \mathbf{J} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{J}) dV$$

$$A \cdot Q = \int_V \rho_v dV$$

$$\int_V (\nabla \cdot \mathbf{J}) dV = -\frac{dQ}{dt} = -\frac{d}{dt} \int_V \rho_v dV$$

$$\boxed{\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}}$$

Unit: A/m^2

DS.1 $\mathbf{J} = 10\rho^2 z \hat{a}_\rho - 4\rho \cos \phi \hat{a}_\phi \text{ mA/m}^2$

(a) Find current density at P ($\rho=3, \phi=30^\circ, z=2$)

$$\begin{aligned} \mathbf{J} &= J_\rho \hat{a}_\rho + J_\phi \hat{a}_\phi + J_z \hat{a}_z \\ &= 10(3)^2(2) \hat{a}_\rho - 4(3) \cos(30^\circ) \hat{a}_\phi \\ &= 180 \hat{a}_\rho - 9 \hat{a}_\phi \text{ mA/m}^2 \end{aligned}$$

(b) Current = ? $\rho=3, 0 < \phi < 2\pi, 2 < z < 2.8$

$$\begin{aligned} I &= \int_V \mathbf{J} \cdot d\mathbf{S} \\ &= \int_2^{2.8} \int_0^{2\pi} \int_0^{2\pi} (10\rho^2 z \hat{a}_\rho - 4\rho \cos \phi \hat{a}_\phi) \cdot \rho d\phi dz \hat{a}_\rho \\ &= \int_2^{2.8} \int_0^{2\pi} 10\rho^2 z \rho d\phi dz \\ &= \int_2^{2.8} 10\rho^3 (2\pi) dz \\ &= 10(27)(2\pi)(0.8) = 3.26 \text{ A} \end{aligned}$$

5.3 Metallic Conductors

- Materials that allow the free flow of electric charges, enabling the conduction of electricity
- There are different energy levels within the atom — electron transition take place b/w these energy levels.
 $\rightarrow$ represent by $K, L, M, N, \dots$

Valence band: The band containing the outermost electrons

Conduction band: Higher energy levels where free electron movement occurs.

Conductor: No gap b/w valence and conduction band
Electrons move freely

Semi-Conductor: A small energy gap exists b/w valence and conduction bands

Small external energies enable conduction

~~Free~~ Insulator: Large energy gap b/w valence and conduction band.

Conduction in Metallic Conductors

$$F = qE$$

$$\Rightarrow F = -eE$$

$$\therefore v = -e$$

Current density

$$J = -\rho_e \mu_e E$$

ρ_e : free electron density

J : current density ($\frac{A}{m^2}$)

drift velocity

$$\Rightarrow v_d = -\mu_e E$$

μ_e : mobility of electrons (measured in $\frac{m^2}{V}$)

$$\Rightarrow \rho = \frac{1}{\sigma}$$

$$\Rightarrow \text{resistivity} = \frac{1}{\text{conductivity}}$$

Ohm's Law:

$$J = \sigma E$$

$$\sigma = \rho_e \mu_e$$

σ : conductivity ($\frac{S}{m}$, Siemens per meter)

$$V = IR$$

$$\Rightarrow R = \frac{L}{\sigma S}$$

$$V = \left(\frac{L}{\sigma S} \right) I$$

$$\Rightarrow V = \left(\frac{L}{\rho_e \mu_e S} \right) I$$

For non-uniform fields,

$$R = - \frac{\int_a^b E \cdot dL}{\int_s \sigma E \cdot ds}$$

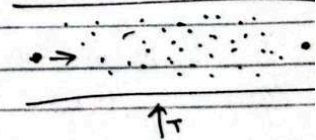
Effect of Temperature

- Increases lattice vibrations

- Decrease mobility μ_e

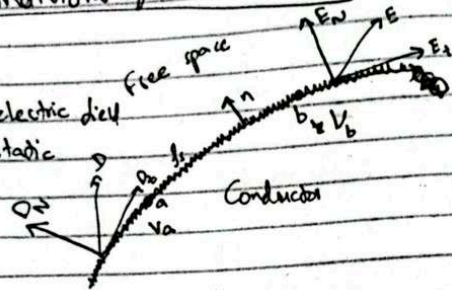
- Decrease conductivity (σ) ($\sigma \propto \mu_e$)

- Increases R ($R \propto \frac{1}{\sigma}$)



Boundary Conditions of a Conductor

- ① $E_t = 0$ → tangential electric field
If $E_t \neq 0$ → violating the static condition



- ② $E_n E_0 = \rho_s$

$$E_n = \frac{\rho_s}{\epsilon_0}$$

∴ No charge exists inside the conductor under static conditions

- ③ Electric flux density

$$D = \epsilon E$$

$$D_n = \frac{\rho_s}{\epsilon_0} \epsilon_0$$

$$D_n = \rho_s$$

∴ The electric field inside a conductor is zero in static conditions.

$$D_t = E_t \epsilon_0 = (0) \epsilon_0$$

$$D_t = 0$$

Equipotential surface
 $V_a = V_b$

5.5 Method of Images

The technique simplifies the solution of problems involving conductors and charge configurations. This method replaces a conductor by an equivalent system of charges that reproduces the electric field in the region of interest.

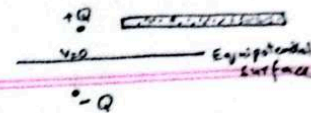
Concept of Images:

- ① In a dipole field, a plane at zero potential b/w the two charges can be represented as an infinite conducting plane.

- Instead of analyzing the system with the conducting plane, we can replace it with a mirror image of charges on the other side of the plane.
- The image charge is opposite in magnitude and located symmetrically wrt conducting plane

point charge Q → its image charge $-Q$

$$V = \frac{Q}{4\pi\epsilon_0|r-r_1|} + \frac{-Q}{4\pi\epsilon_0|r-r_2|}$$

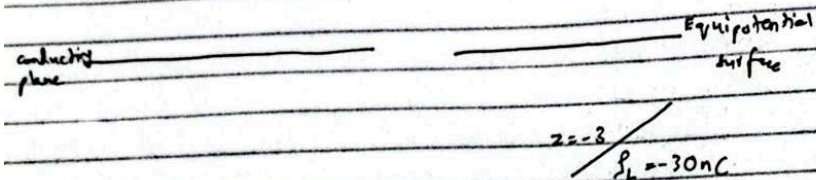


Q) $\rho_L = 30 \text{ nC/m}$ at $z=3 \text{ m}$ along the $x=0$ axis

- The conducting plane lies at $z=0$

- Find E at $P(2,5,0)$

So:



Replace the conducting plane with an image line charge of $\rho_L = -30 \text{ nC/m}$ at $z=-3$.

From +ive line charge to $P(2,5,0)$: $\vec{R}_+ = 2\vec{a}_x - 3\vec{a}_z$

From -ive line charge to $P(2,5,0)$: $\vec{R}_- = 2\vec{a}_x + 3\vec{a}_z$

$$\vec{E}_+ = \frac{\rho_L}{2\pi\epsilon_0 R_+^2} \frac{\vec{R}_+}{|\vec{R}_+|}$$

$$\vec{E}_- = \frac{-\rho_L}{2\pi\epsilon_0 R_-^2} \frac{\vec{R}_-}{|\vec{R}_-|}$$

$$\vec{E} = \vec{E}_+ + \vec{E}_- = -249 \vec{a}_z \text{ V/m}$$

$$\vec{D} = \vec{E}\epsilon_0 = -2.20 \vec{a}_z \text{ nC/m}^2 = \rho_s$$

Ch#5

Remaining Online Lecture

Classroom Pdf

Cn#06

Capacitance

Capacitance is a measure of a system's ability to store electric charge. It is defined in terms of two oppositely charged conductors separated by a dielectric medium.

- Electric field b/w conductor is normal to surface
- Each conductor is an equipotential surface. The potential difference V_0 exists b/w two conductors.

Capacitance 'C' is defined as the ratio of the total charge on either conductor to the potential difference b/w

$$C = \frac{Q}{V_0}$$

→ Q: Determined by the V_0 surface integration over the conductor.

→ V_0 : Calculated by integrating the electric field along a path from -ive to +ive.

$$C = \frac{\int_S \mathbf{E} \cdot d\mathbf{S}}{-\int_+^- \mathbf{E} \cdot d\mathbf{L}}$$

Units:

SI-unit of capacitance: Farad (F)

6.2 Parallel Plate Capacitor

Consisting of two parallel conducting plates separated.

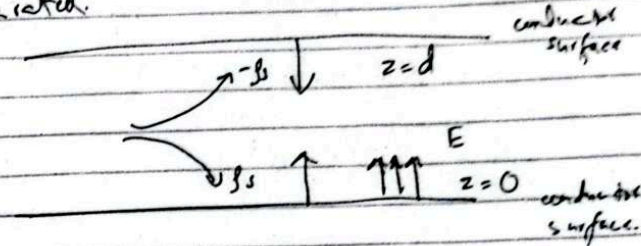


Plate separation: d
Plate area: S

Dielectric permittivity: ϵ
Electric flux density: $\mathbf{D} = \rho_s \hat{\mathbf{a}}_z$

Energy stored in capacitor

$$W_E = \frac{1}{2} CV^2 = \frac{1}{2} QV_0 = \frac{1}{2} \frac{Q^2}{C}$$

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{\rho_s}{\epsilon} \hat{\mathbf{a}}_z$$

$\epsilon_r \rightarrow \epsilon \rightarrow \infty$
than air
increased

$$V_0 = - \int_d^0 \mathbf{E} \cdot d\mathbf{L} = - \int_d^0 \frac{\rho_s}{\epsilon} \cdot dL$$

$$V_0 = \frac{\rho_s d}{\epsilon} \quad (\text{-ive sign is accordance to reference pt})$$

$$C = \frac{\epsilon S}{d}$$

$$Q = \rho_s S$$

$$C = \frac{Q}{V} = \frac{\rho_s S}{\frac{\rho_s d}{\epsilon}} = \frac{\epsilon S}{d}$$

$$C = \frac{\epsilon S}{d} \rightarrow \text{area} \rightarrow \text{separation b/w plates}$$

Example)

$$\epsilon_r = 6, A = 10 \text{ m}^2, d = 0.01, C = ?$$

$$\Rightarrow C = \frac{\epsilon S}{d} = \frac{\epsilon_0 \epsilon_r S}{d}$$

$$= \frac{(8.854 \times 10^{-12}) (6) (10)}{0.01}$$

$$C = 5.312 \times 10^{-8} \text{ F}$$

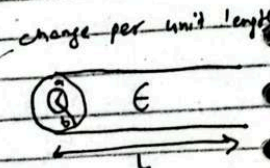
Energy stored in a Capacitor

$$W_E = \frac{1}{2} C V_0^2 = \frac{1}{2} Q V_0 = \frac{1}{2} \frac{Q^2}{C}$$

Capacitance : Use case 1

1) Co-axial Cable

$$\epsilon_r = \frac{Q/L}{2\pi \epsilon_r r} = \frac{\lambda}{2\pi \epsilon_r r}$$



$$V_0 = \int_a^b \epsilon_r dr = \frac{\lambda}{2\pi \epsilon} \ln \frac{b}{a}$$

$$\int_a^b \frac{dr}{r} = \ln \left(\frac{b}{a} \right)$$

$$C = \frac{Q}{V_0} = \frac{2\pi \epsilon L}{\ln \left(\frac{b}{a} \right)}$$

$$V_0 = \frac{\lambda}{2\pi \epsilon} \ln \left(\frac{b}{a} \right)$$

$$= \frac{Q/L}{2\pi \epsilon} \ln \left(\frac{b}{a} \right)$$

$$\frac{Q}{L} = \frac{V_0 \cdot 2\pi \epsilon}{\ln \left(\frac{b}{a} \right)}$$

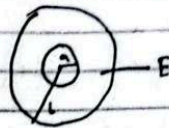
$$C = \frac{Q}{V_0} = \frac{2\pi \epsilon L}{\ln \left(\frac{b}{a} \right)}$$

2) Spherical Capacitor

Inner radius 'a'
Outer radius 'b'
Permittivity 'epsilon'

$$E_r = \frac{Q}{4\pi \epsilon r^2}$$

$$V_0 = \int_a^b E_r dr$$



$$V_0 = \frac{Q}{4\pi \epsilon} \int_a^b \frac{1}{r^2} dr$$

$$= \frac{Q}{4\pi \epsilon} \left[-\frac{2}{r} \right] dr$$

$$= -\frac{Q}{2\pi \epsilon} \left[\frac{1}{b} - \frac{1}{a} \right]$$

invert to remove -ive sign

$$= \frac{Q}{2\pi \epsilon} \left[\frac{1}{a} - \frac{1}{b} \right]$$

$$V_0 = \frac{Q}{2\pi \epsilon} \left[\frac{1}{a} - \frac{1}{b} \right]$$

$$C = \frac{Q}{V_0} = 4\pi \epsilon \left(\frac{ab}{b-a} \right)$$

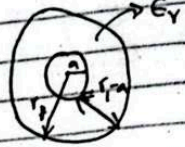
Capacitance depends on inner & outer radius if $b \rightarrow \infty$

$$C = 4\pi \epsilon a$$

1) Spherical Capacitor with dielectric layer

Sphere with radius $\rightarrow a$
dielectric layer thickness $\rightarrow r_1 - a$
Outer conductor radius $\rightarrow r_1$

$$C = \frac{4\pi}{\frac{1}{\epsilon_r} \left(\frac{1}{a} - \frac{1}{r_1} \right) + \frac{1}{\epsilon_0 r_1}}$$

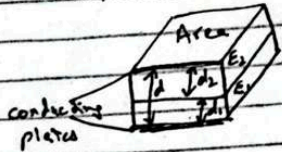


Parallel Plate Capacitor with Multiple Dielectric

$$C = \frac{\epsilon S}{d} \quad \because d = d_1 + d_2$$

$$C_1 = \frac{\epsilon_1 S}{d_1}$$

$$C_2 = \frac{\epsilon_2 S}{d_2}$$



$$\frac{1}{C_{Total}} = \frac{1}{C_1} + \frac{1}{C_2}$$

(capacitance in series)

$$C_{Total} = \frac{1}{\frac{d_1}{\epsilon_1 S} + \frac{d_2}{\epsilon_2 S}} = \frac{1}{\frac{1}{S} \left(\frac{d_1 \epsilon_2 + d_2 \epsilon_1}{\epsilon_1 \epsilon_2} \right)}$$

$$C_{Total} = \frac{S(\epsilon_1 \epsilon_2)}{d_1 \epsilon_2 + d_2 \epsilon_1}$$

2) Coaxial Cable Capacitor

inner conductor diameter = 0.1045 in

Outer " " = 0.620 in

Dielectric = $\epsilon_r = 2.26$

Length = $L = 1$ ft

$C = ?$

$$C = \frac{2\pi \epsilon_0 \epsilon_r L}{\ln \left(\frac{b}{a} \right)}$$

3) Parallel plate Capacitor

$A = 1 \text{ cm} \times 4 \text{ cm} = 4 \text{ cm}^2$

$\epsilon_1 = 1.5, \epsilon_2 = 2.5, \epsilon_3 = 6$

$d_1 = d_2 = d_3 = 0.1 \text{ mm}$

$C_1 = ?$, $C_2, C_3 = ?$

$$C_1 = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)^{-1}$$

$$\because C = \frac{\epsilon S}{d}$$

$$= \left(\frac{d}{\epsilon_1 S} + \frac{d}{\epsilon_2 S} + \frac{d}{\epsilon_3 S} \right)^{-1}$$

$$= \left[\frac{d}{S} \left(\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} + \frac{1}{\epsilon_3} \right) \right]^{-1}$$

$$= 28.6 \text{ pF}$$

Laplace Equations/Poisson Equation

Gauss's law:

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho_v \\ \mathbf{D} &= \epsilon \mathbf{E} \\ \mathbf{E} &= -\nabla V \\ \mathbf{D} &= \epsilon (-\nabla V)\end{aligned}$$

$$\nabla \cdot (-\nabla V) \epsilon = \rho_v$$

$$\boxed{\nabla \cdot \nabla V = -\frac{\rho_v}{\epsilon}} \rightarrow \text{Poisson Eq.}$$

$$\nabla V = \frac{\partial V}{\partial x} \hat{a}_x + \frac{\partial V}{\partial y} \hat{a}_y + \frac{\partial V}{\partial z} \hat{a}_z$$

$$\nabla \cdot \nabla V = \frac{\partial^2 V}{\partial x^2} \hat{a}_x + \frac{\partial^2 V}{\partial y^2} \hat{a}_y + \frac{\partial^2 V}{\partial z^2} \hat{a}_z = -\frac{\rho_v}{\epsilon}$$

If $\rho_v = 0 \rightarrow$ zero volume charge density
but there is a possibility $\rightarrow$ its a point
charge on line charge, then

$$\nabla \cdot \nabla V = \boxed{\nabla^2 V = 0} \rightarrow \text{Laplace eq.}$$

Q) Calculate V and ρ_v at point 'P'

a) $V = \frac{4y^2}{x^2+1}$ at $P(1,2,3)$

$$V = \frac{4(2)(3)}{(1)^2+1} = 12$$

$$V_x = \frac{4y^2}{x^2+1}$$

$$\begin{aligned}V_x' &= 4y^2 (x^2+1)^{-2} \\ &= -4y^2 (x^2+1)^{-2} (2x) = -8xy^2 (x^2+1)^{-2}\end{aligned}$$

Steps:-

- 1) Given V , use $\mathbf{E} = -\nabla V$, find $\mathbf{E}$
- 2) Use $\mathbf{D} = \epsilon \mathbf{E}$ to find $\mathbf{D}$
- 3) Evaluate $\mathbf{D}$ at either capacitor plate
 $\mathbf{D} = \mathbf{D}_S = \mathbf{D}_N \text{ on}$
- 4) $\rho_s = \mathbf{D}_N$
- 5) Find Q by surface integration over
the capacitor $Q = \int_S \rho_s dS$
- 6) $C = \frac{Q}{V}$

$\Rightarrow$ Parallel Plate Capacitor of Area S and potential
difference V_0 b/w them

Laplace eq. $\frac{\partial^2 V}{\partial u^2} = 0$

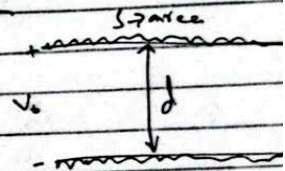
$$\frac{dV}{du} = A$$

$$V = Ax + B \rightarrow 0$$

$$A = \frac{V_0}{d} \rightarrow \text{slope}$$

intercept $\Rightarrow B = 0$ put in 0

$$\boxed{V = \frac{V_0 x}{d}}$$



$$① E = -\nabla V = -\frac{V_0}{d} \hat{a}_n$$

$$② D = -\epsilon \frac{V_0}{d} \hat{a}_n$$

$$③ D_s = D|_{x=0} \rightarrow \text{There is no axis other than 'x' and point lie on x-axis. So } D_s = -\epsilon \frac{V_0}{d} \hat{a}_n$$

$$④ \oint_s D_s = -\epsilon \frac{V_0}{d} \hat{a}_n$$

$$⑤ Q = \int_s -\frac{\epsilon V_0}{d} dS = -\epsilon \frac{V_0}{d} \int_s dS = -\epsilon \frac{V_0 S}{d}$$

$$⑥ C = \frac{|Q|}{V_0} = \frac{\epsilon_0 S V_0}{d V_0} = \frac{\epsilon S}{d}$$

$$C = \frac{\epsilon S}{d}$$

D6.6) Find $|E|$ at $P(3, 2)$ in RCS for field of

① two coaxial conducting cylinder $V=50V$ at $\rho=2m$ and $V=20$ at $\rho=3m$

② Two radial conducting $V=50$ at $\phi=10^\circ$ and $V=20V$ at $\phi=30^\circ$

Sol.

For co-axial cylinders

$$V(\rho) = \frac{-A}{\ln(\frac{\rho}{a})} + C$$

$$a=2m \xrightarrow{V(a)} 50V \quad 50 = -\frac{A}{\ln(\frac{2}{a})} + C$$

$$b=3m \xrightarrow{V(b)} 20V$$

$$V(\rho) = \quad 20 = \frac{-A}{\ln(\frac{3}{2})} + C \quad \text{--- ②}$$

Comparing ① & ②

$$A = 30$$

$$\text{So } C = 80$$

$$V(\rho) = -30 \ln \rho + 80$$

$$E = -\nabla V = -\frac{\partial V}{\partial \rho} = -\frac{\partial}{\partial \rho} (-30 \ln \rho + 80)$$

$$= \frac{30}{\rho}$$

$$\rho = \sqrt{(3)^2 + (2)^2} = \sqrt{13}$$

$$E = \frac{30}{\sqrt{13}}$$

Ch # 7

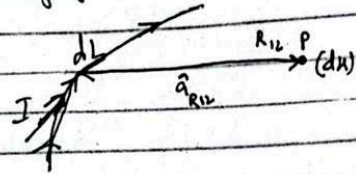
Bio-Savart's Law

Fundamentals of Law in Electromagnetism:

Sources of magnetic field:

- 1) Permanent magnet
- 2) Electric field varying with time
- 3) Direct current

Magnetic field Intensity
'H' → Unit: A/m



$$dH = \frac{IdL \times \hat{a}_{R12}}{4\pi R_{12}^2}$$

(direction: right hand
thumb rule)

$$dH = \frac{IdL \times R_{12}}{4\pi R_{12}^3}$$

for continuous current
path

$$H = \int \frac{IdL \times \hat{a}_r}{4\pi R^2}$$

$$dH \propto I$$

$$dH \propto dL$$

$$dH \propto \frac{1}{R^2}$$

Analogue with Coulomb's Law:-

Coulomb's Law	Bio-Savart Law
→ Field produced by charge element	→ Field produced by current element
→ Inverse square dependence on distance	→ Inverse square dependence on distance
→ Electric field is radial	→ Magnetic field is tangential

Surface Charge Density (2)

→ for a sheet of current

$$H = \int \frac{K \times \hat{a}_r}{4\pi R^2} dS \quad \left(\frac{A/m \times m^2}{m^2} \right)$$

Unit of K is A/m

→ Volume Current Density J
 for volume distribution

$$H = \int \frac{J \times \hat{a}_r}{4\pi R^2} dV$$

Unit of J is A/m²

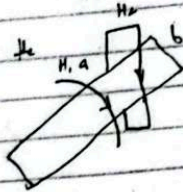
⇒ Magnetic field of an Infinite straight Filament

$$H = \frac{I}{2\pi \rho} \hat{a}_\phi$$

7.2 Ampere's Circuital Law

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{\text{enclosed}}$$

Direction of current using the right hand rule.
Analogue with Gauss's law



$$\oint \mathbf{E} \cdot d\mathbf{A} = Q_{\text{enclosed}} / \epsilon_0$$

Applications:

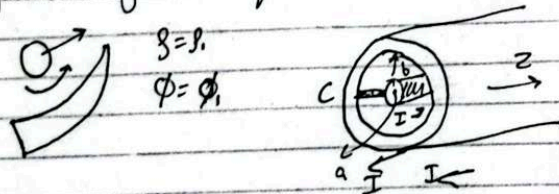
- ① Magnetic field of an infinite element carrying element / filament

$$H = \frac{I}{2\pi r}$$

$$\oint \mathbf{H} \cdot d\mathbf{L} = I$$

by symmetry 'H' has no 'z-axis' dependence.

- ② Magnetic field of a Co-axial Cable



- ③ If $r < a$

(inside the inner conductor)

$$H\phi = \frac{I r}{2\pi a^2}$$

- ④ $a < r < b$ (between conductors)

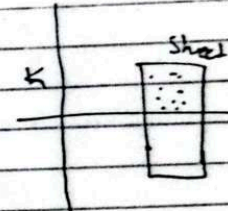
$$H\phi = \frac{I}{2\pi r}$$

- ⑤ $r > b$

$$H\phi = 0$$

- ⑥ Magnetic field of a Current sheet

Surface current density K

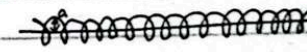


$$H = \frac{1}{2} K \text{ (above the sheet } z > 0)$$

$$H = -\frac{1}{2} K \text{ (below the sheet } z < 0)$$

- ⑦ Infinite Solenoid

radius of solenoid is 'a'



surface charge density = $K a \phi$

H: inside the solenoid and outside the solenoid

$$\text{Inside: } H = K a \phi$$

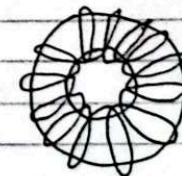
$$\text{Outside: } H = 0$$

- ⑧ Magnetic field of a Toroid

A toroidal coil with 'N' turns, radius 'r'.

$$\text{Inside: } H = \frac{NI}{2\pi r} \hat{a}_\phi$$

$$\text{Outside: } H = 0$$



$$z=80$$

$$z=01$$

1) Find H at $P(0, 0.2, 0)$ in the field of

(a) Current filament $2.5A$ in direction $\hat{a}_z$ at

$$x=0.1, y=0.3$$

(b) Co-axial cable centered on z -axis with

$$a=0.3, b=0.5, c=0.6, I=2.5A$$

(c) Three current sheets $2.7 \hat{a}_x$ A/m at

$$y=0.1, -1.4 \hat{a}_x \text{ A/m at } y=0.15 \text{ and}$$

$$-1.3 \hat{a}_x \text{ A/m at } y=0.25$$

1) (a) Position vector between filament and P :

$$\vec{R} = \vec{r}_P - \vec{r}_{\text{filament}}$$

$$= (0-0.1)\hat{a}_x - (0.2-0.3)\hat{a}_y$$

$$= -0.1\hat{a}_x - (-0.1)\hat{a}_y$$

$$= -0.1\hat{a}_x + 0.1\hat{a}_y$$

$$H = \frac{I}{2\pi R} \hat{R} = \frac{I}{2\pi R^2} \vec{R}$$

$$= \frac{-2.5}{2\pi (0.1414)^2} \vec{R} \left\{ |\vec{R}| \sqrt{(-0.1)^2 + (0.1)^2} \right. \\ \left. = 0.1414 \right.$$

$$H = -1.989 \hat{a}_x + 1.989 \hat{a}_y \text{ A/m}$$

(b) $a=0.3m, b=0.5m, c=0.6m$

$$I=2.5A, P(0, 0.2, 0) \quad p=|P|=0.2$$

$$p < a$$

$$H = \frac{I p}{2\pi a^2} = \frac{(2.5)(0.2)}{2\pi (0.3)^2} = 0.844 \text{ A/m}$$

$$H\hat{a} = 0.844 \hat{a}_x \text{ A/m}$$

(c) Three sheets:

$$y=0.1, K=2.7 \hat{a}_z \text{ A/m}$$

$$y=0.15, K=-1.4 \hat{a}_z \text{ A/m}$$

$$y=0.25, K=-1.3 \hat{a}_z \text{ A/m}$$

Point $P(0, 0.2, 0) \quad p_y=0.2$

$$\left. \begin{aligned} y=0.1 &\Rightarrow H = \frac{1}{2} K = \frac{1}{2} (2.7) = 1.35 \hat{a}_z \text{ A/m} \\ y=0.15 &\Rightarrow H = \frac{1}{2} (-1.4) = -0.7 \hat{a}_z \text{ A/m} \\ y=0.25 &\Rightarrow H = \frac{1}{2} (-1.3) = -0.65 \hat{a}_z \text{ A/m} \end{aligned} \right\} \begin{array}{l} \text{Sum} \\ \text{final} \\ H = 1.35 \hat{a}_z \text{ A/m} \end{array}$$